PRACTICAL 2

NAME: Salihundam Svara Haasini

TASKS :

a) Creating First Linux Instance

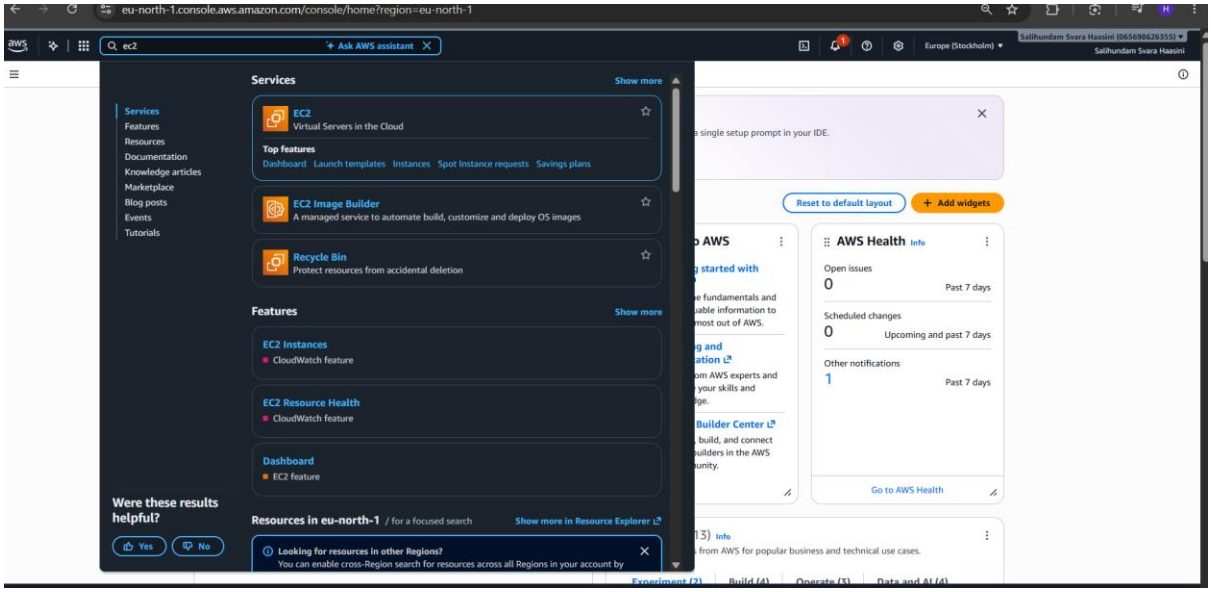
b) S3 Bucket Creation and Upload Objects

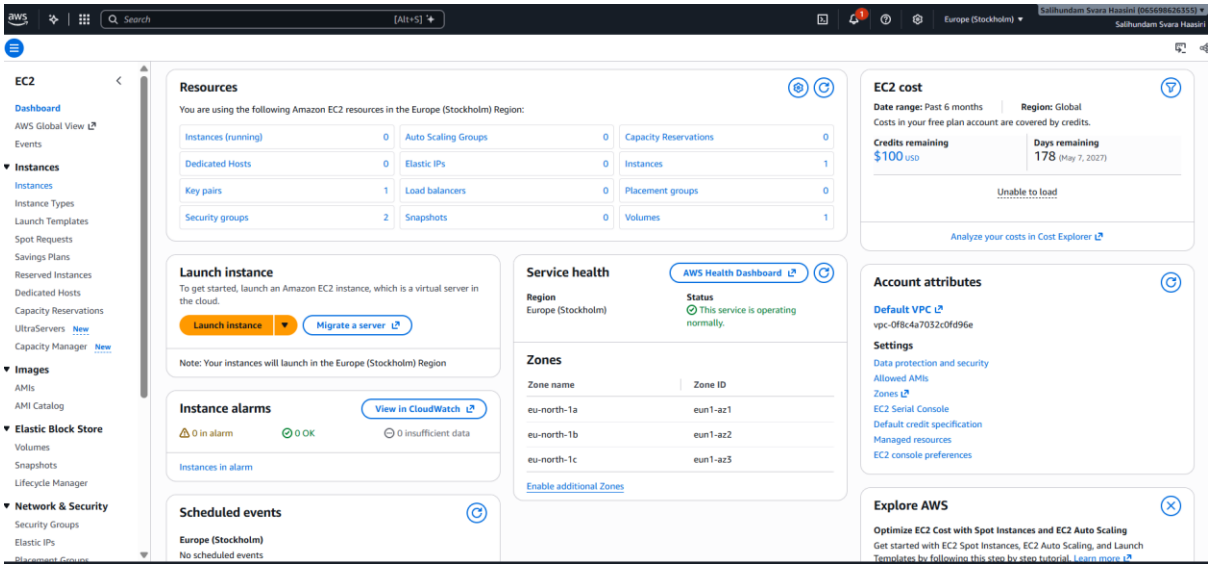
I.Create an S3 Bucket for file storage.

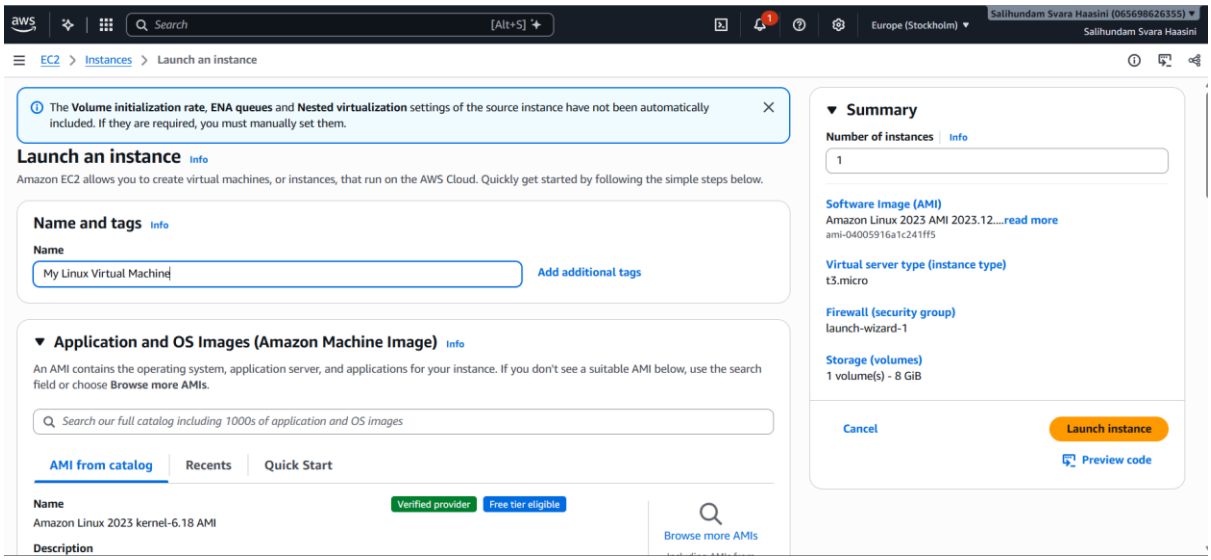
II.Upload 5 objects with different file extensions.

TASK : a) Creating First Linux Instance

i. Creating a Linux Virtual Machine




Search
[Alt+S]
Europe (Stockholm)
Salihundam Svara Haasini (065698626355)
Salihundam Svara Haasini
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Resources
You are using the following Amazon EC2 resources in the Europe (Stockholm) Region:
Instances (running) 0
Auto Scaling Groups 0
Capacity Reservations 0
Dedicated Hosts 0
Elastic IPs 0
Instances 1
Key pairs 1
Load balancers 0
Placement groups 0
Security groups 2
Snapshots 0
Volumes 1
EC2 cost
Date range: Past 6 months
Region: Global
Costs in your free plan account are covered by credits.
Credits remaining
$100 USD
Days remaining
178 (May 7, 2027)
Unable to load
Analyze your costs in Cost Explorer
Launch instance
To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.
Launch instance
Migrate a server
Note: Your instances will launch in the Europe (Stockholm) Region
Service health
AWS Health Dashboard
Region
Europe (Stockholm)
Status
This service is operating normally.
Zones
Zone name
Zone ID
eu-north-1a
eun1-az1
eu-north-1b
eun1-az2
eu-north-1c
eun1-az3
Enable additional Zones
Account attributes
Default VPC
vpc-0f8c4a7032c0fd96e
Settings
Data protection and security
Allowed AMIs
Zones
EC2 Serial Console
Default credit specification
Managed resources
EC2 console preferences
Instance alarms
View in CloudWatch
0 in alarm
0 OK
0 insufficient data
Instances in alarm
Scheduled events
Europe (Stockholm)
No scheduled events
Explore AWS
Optimize EC2 Cost with Spot Instances and EC2 Auto Scaling
Get started with EC2 Spot Instances, EC2 Auto Scaling, and Launch Templates by following this step by step tutorial. Learn more


Search
[Alt+S]
Europe (Stockholm)
Salihundam Svara Haasini (065698626355)
Salihundam Svara Haasini
EC2 > Instances > Launch an instance
The Volume initialization rate, ENA queues and Nested virtualization settings of the source instance have not been automatically included. If they are required, you must manually set them.
Launch an instance Info
Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.
Name and tags Info
Name
My Linux Virtual Machine
Add additional tags
Application and OS Images (Amazon Machine Image) Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
AMI from catalog
Recents
Quick Start
Name
Amazon Linux 2023 kernel-6.18 AMI
Verified provider
Free tier eligible
Description
Browse more AMIs
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....read more
ami-04005916a1c241ff5
Virtual server type (instance type)
t3.micro
Firewall (security group)
launch-wizard-1
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code

Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Search our full catalog including 1000s of application and OS images

Recents | Quick Start

Amazon Linux | macOS | Ubuntu | Windows | Red Hat | SUSE Linux | Debian

Browse more AMIs
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.18 AMI — Free tier eligible
ami-04005916a1c241ff5 (64-bit (x86), uefi-preferred) / ami-0073641137e019ef3 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Amazon Linux 2023 (kernel-6.18) is a modern, general purpose Linux-based OS that will be supported until June 2029. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.12.20261109.0 x86_64 HVM kernel-6.18

Instance type Info | Get advice

Instance type

t3.micro — Free tier eligible
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0108 USD per Hour On-Demand Windows base pricing: 0.02 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0143 USD per Hour
On-Demand RHEL base pricing: 0.0396 USD per Hour On-Demand SUSE base pricing: 0.0108 USD per Hour

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

Proceed without a key pair (Not recommended) — Default value

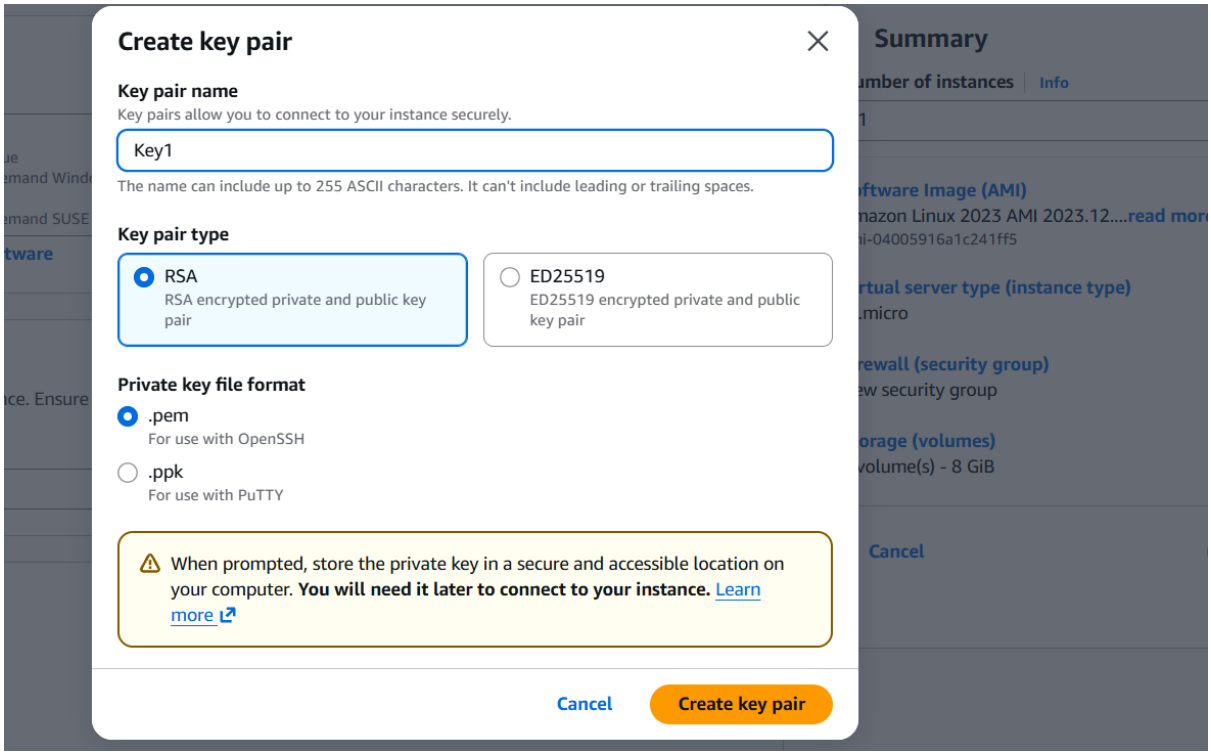
Create new key pair

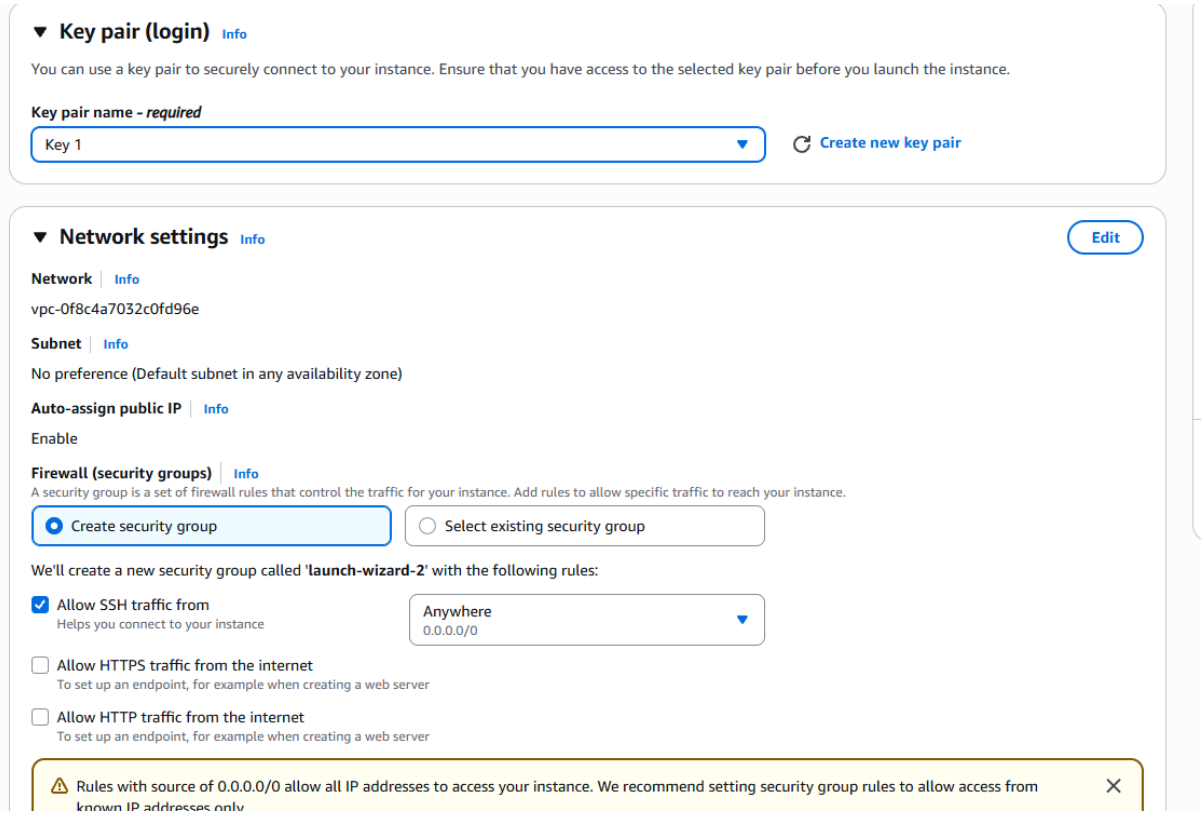
Network settings Info

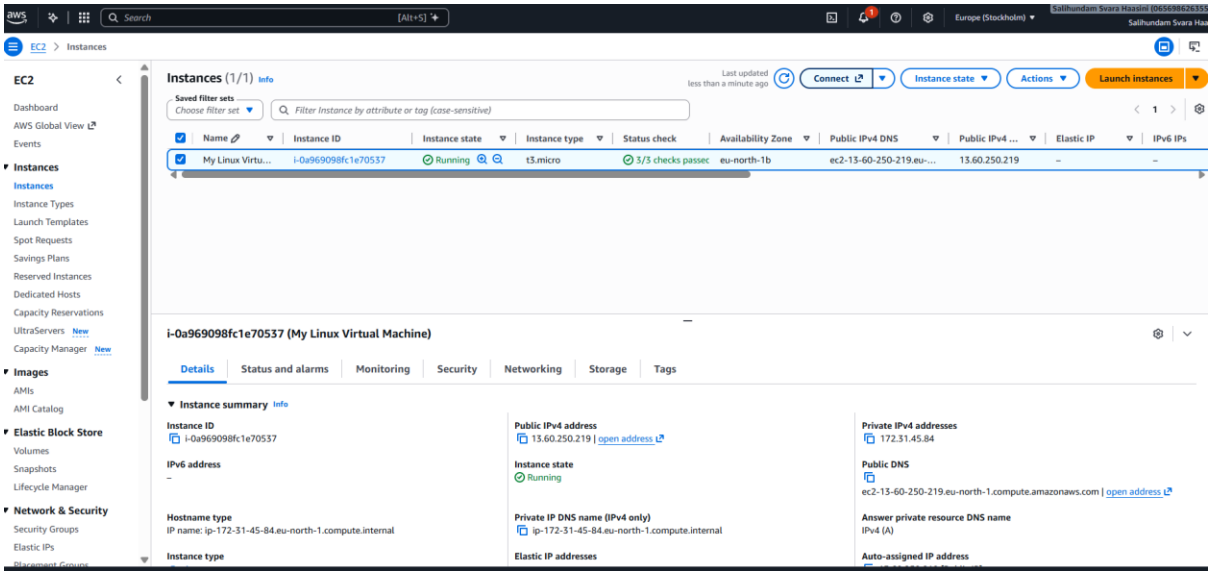
Edit

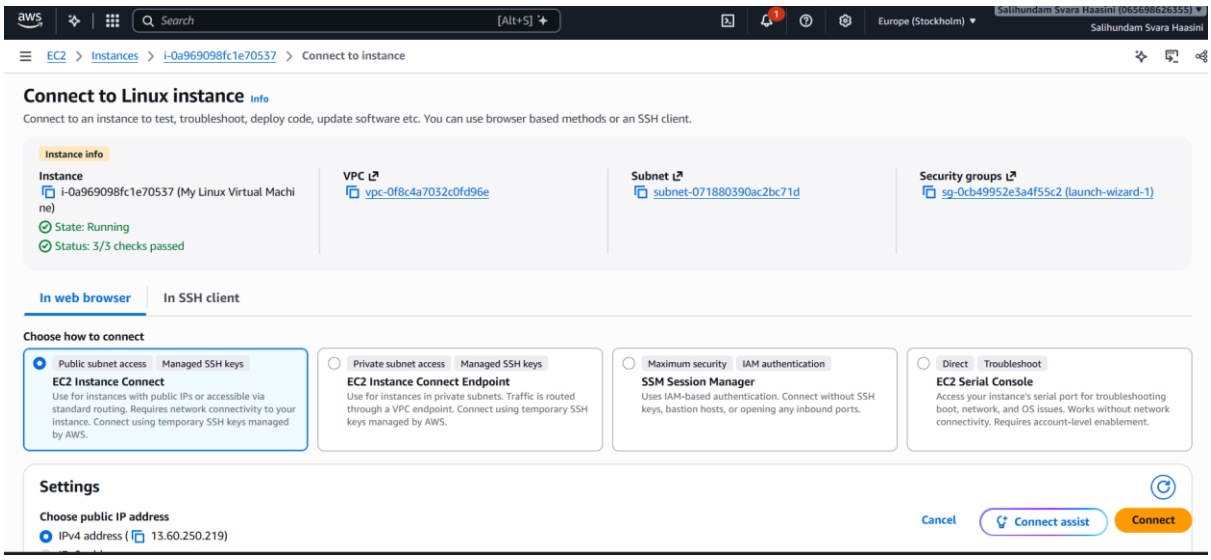
Network | Info

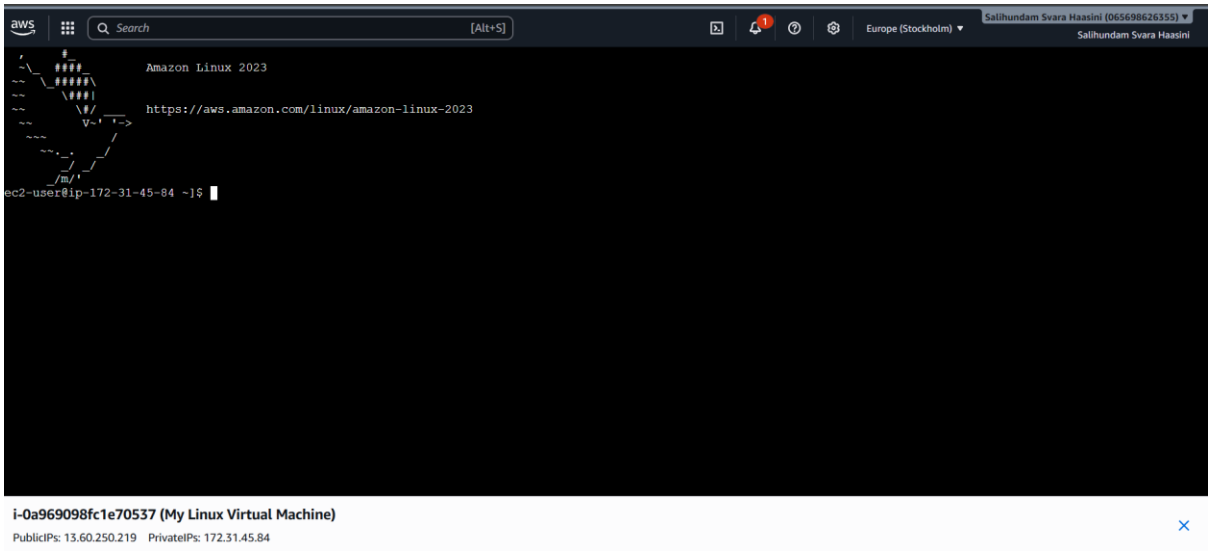
vpc-0f8c4a7032c0fd96e


Create key pair
Key pair name
Key pairs allow you to connect to your instance securely.
Key1
The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.
Key pair type
RSA
RSA encrypted private and public key pair
ED25519
ED25519 encrypted private and public key pair
Private key file format
.pem
For use with OpenSSH
.ppk
For use with PuTTY
When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. Learn more
Cancel
Create key pair
Summary
Cancel

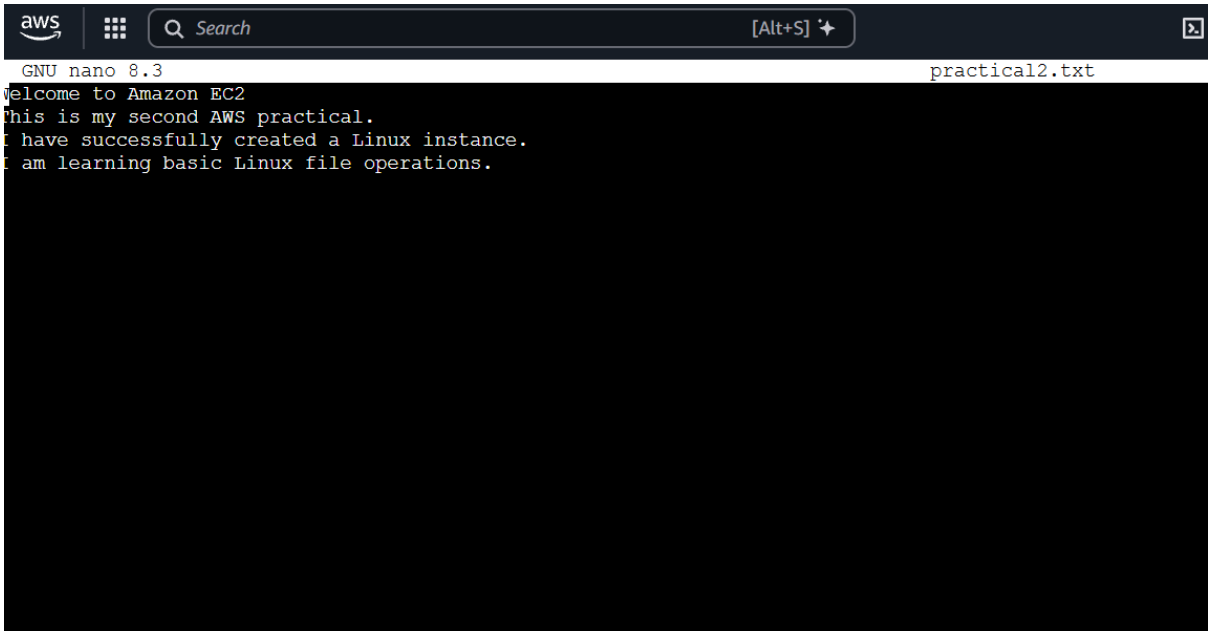

Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
Key 1
Create new key pair
Network settings Info
Edit
Network Info
vpc-0f8c4a7032c0fd96e
Subnet Info
No preference (Default subnet in any availability zone)
Auto-assign public IP Info
Enable
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
We'll create a new security group called 'launch-wizard-2' with the following rules:
Allow SSH traffic from
Helps you connect to your instance
Anywhere
0.0.0.0/0
Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server
Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server
Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only


Search
[Alt+S]
Europe (Stockholm)
Salihundam Svara Haasini
EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Instances (1/1) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
Public IPv4 DNS
Public IPv4 ...
Elastic IP
IPv6 IPs
My Linux Virtu...
i-0a969098fc1e70537
Running
t3.micro
3/3 checks passed
eu-north-1b
ec2-13-60-250-219.eu-...
13.60.250.219
i-0a969098fc1e70537 (My Linux Virtual Machine)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-0a969098fc1e70537
Public IPv4 address
13.60.250.219 | open address
Private IPv4 addresses
172.31.45.84
IPv6 address
Instance state
Running
Public DNS
ec2-13-60-250-219.eu-north-1.compute.amazonaws.com | open address
Hostname type
IP name: ip-172-31-45-84.eu-north-1.compute.internal
Private IP DNS name (IPv4 only)
ip-172-31-45-84.eu-north-1.compute.internal
Answer private resource DNS name
IPv4 (A)
Instance type
Elastic IP addresses
Auto-assigned IP address


Search
[Alt+S]
Europe (Stockholm)
Salihundam Svara Haasini (065698626355)
Salihundam Svara Haasini
EC2 > Instances > i-0a969098fc1e70537 > Connect to instance
Connect to Linux instance Info
Connect to an instance to test, troubleshoot, deploy code, update software etc. You can use browser based methods or an SSH client.
Instance info
Instance
i-0a969098fc1e70537 (My Linux Virtual Machine)
State: Running
Status: 3/3 checks passed
VPC
vpc-0f8c4a7032c0fd96e
Subnet
subnet-071880390ac2bc71d
Security groups
sg-0cb49952e3a4f55c2 (launch-wizard-1)
In web browser
In SSH client
Choose how to connect
Public subnet access
Managed SSH keys
EC2 Instance Connect
Use for instances with public IPs or accessible via standard routing. Requires network connectivity to your instance. Connect using temporary SSH keys managed by AWS.
Private subnet access
Managed SSH keys
EC2 Instance Connect Endpoint
Use for instances in private subnets. Traffic is routed through a VPC endpoint. Connect using temporary SSH keys managed by AWS.
Maximum security
IAM authentication
SSM Session Manager
Uses IAM-based authentication. Connect without SSH keys, bastion hosts, or opening any inbound ports.
Direct
Troubleshoot
EC2 Serial Console
Access your instance's serial port for troubleshooting boot, network, and OS issues. Works without network connectivity. Requires account-level enablement.
Settings
Choose public IP address
IPv4 address (13.60.250.219)
Cancel
Connect assist
Connect




```
[ec2-user@ip-172-31-45-84 ~]$ mkdir AWS_Practical
[ec2-user@ip-172-31-45-84 ~]$ cd AWS_Practical
[ec2-user@ip-172-31-45-84 AWS_Practical]$ pwd
/home/ec2-user/AWS_Practical
```

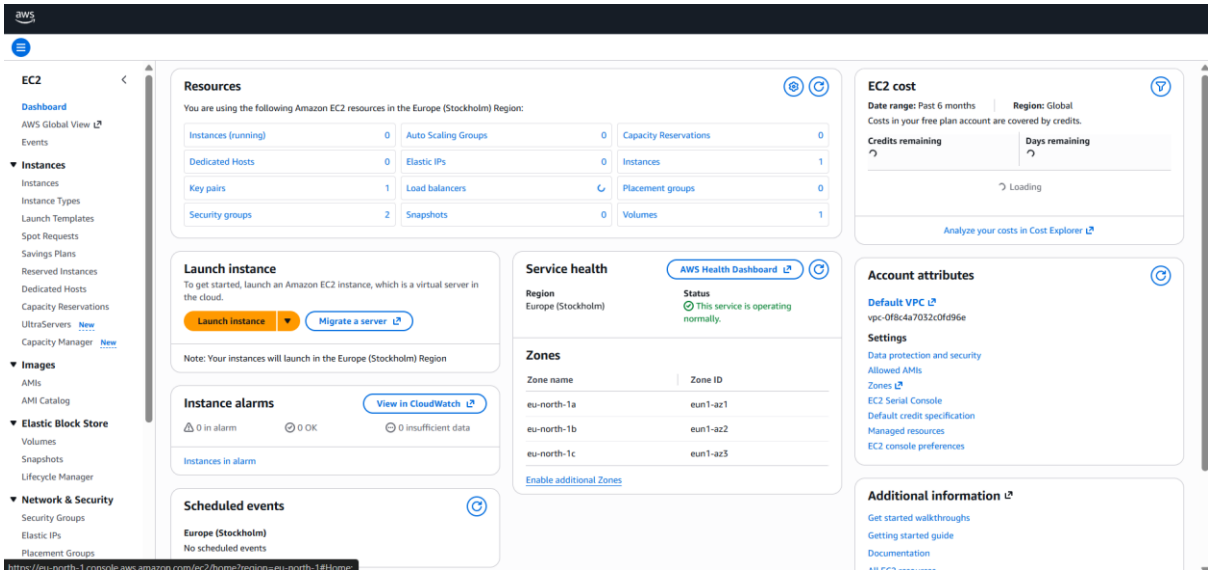


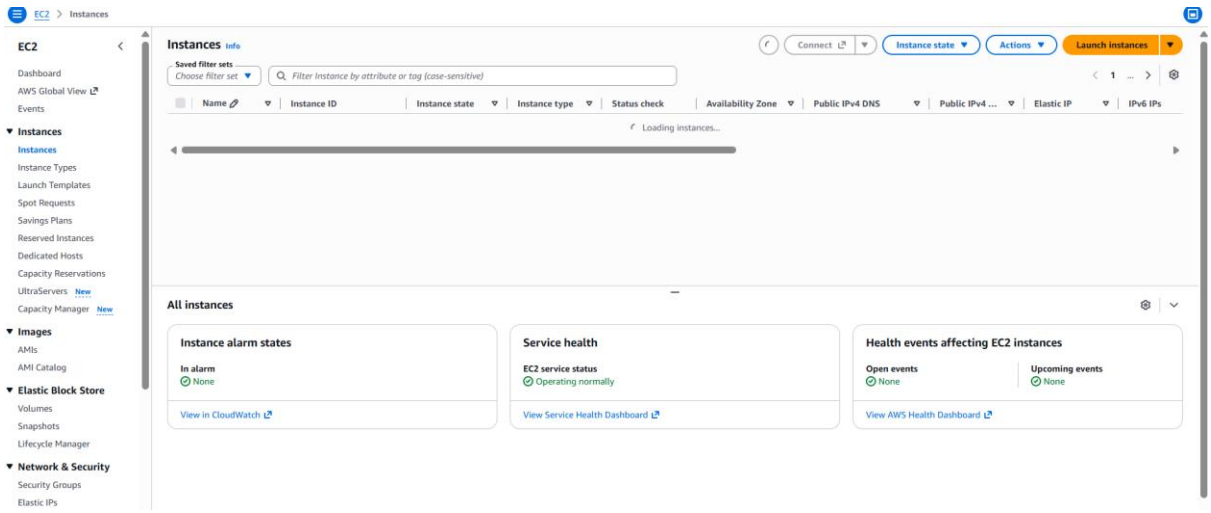
```
[ec2-user@ip-172-31-45-84 AWS_Practical]$ pwd
/home/ec2-user/AWS_Practical
[ec2-user@ip-172-31-45-84 AWS_Practical]$ ls
practical2.txt
[ec2-user@ip-172-31-45-84 AWS_Practical]$ nano practical2.txt
[ec2-user@ip-172-31-45-84 AWS_Practical]$ cat practical2.txt
Welcome to Amazon EC2
This is my second AWS practical.
I have successfully created a Linux instance.
I am learning basic Linux file operations.
[ec2-user@ip-172-31-45-84 AWS_Practical]$
```

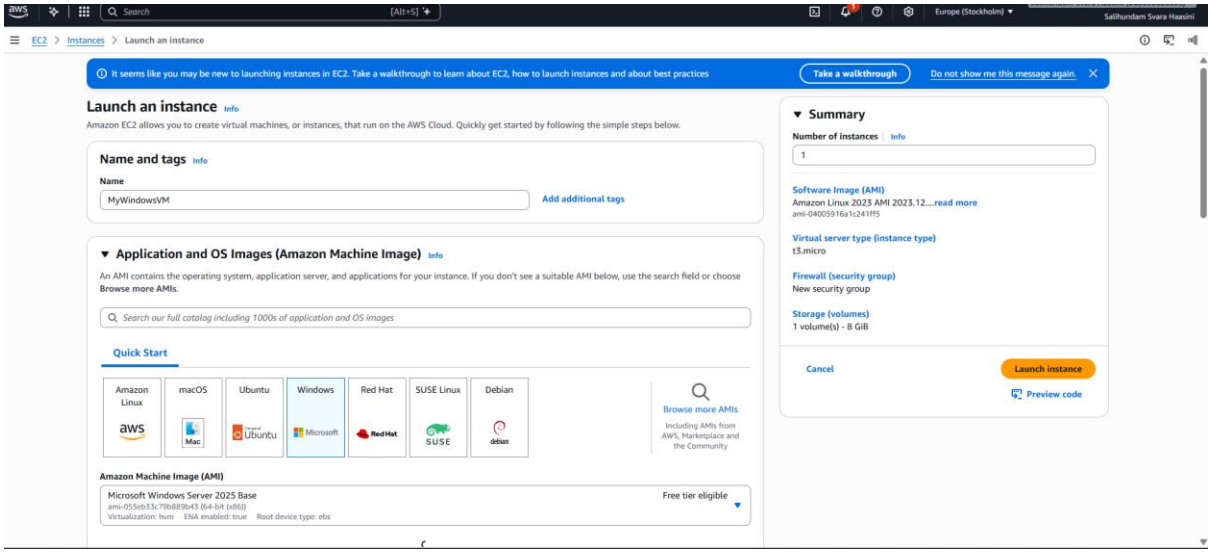
i-0a969098fc1e70537 (My Linux Virtual Machine)

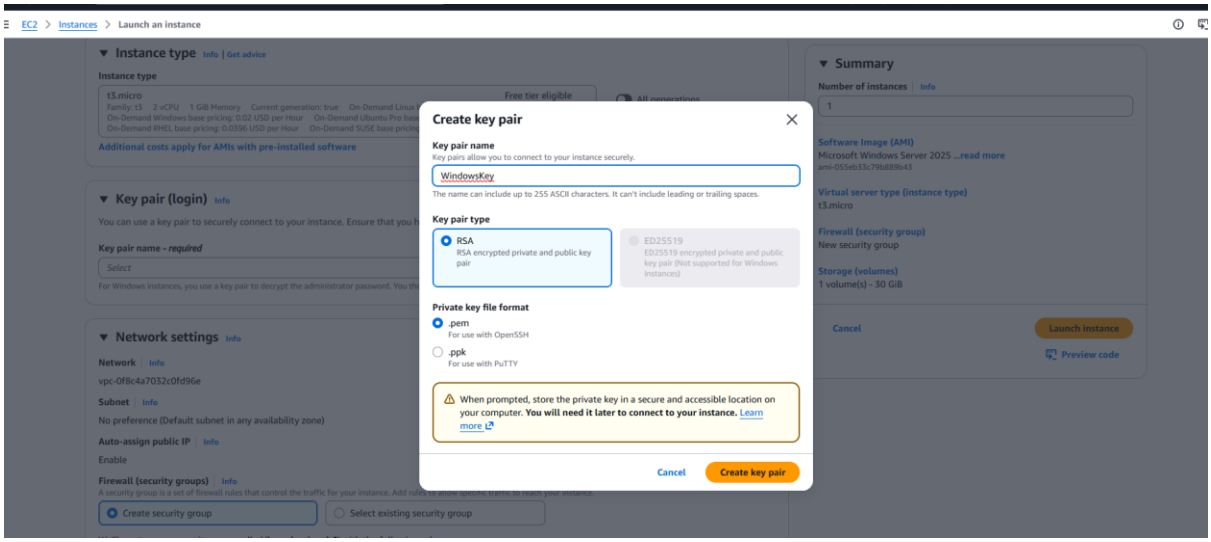
PublicIPs: 13.60.250.219 PrivateIPs: 172.31.45.84

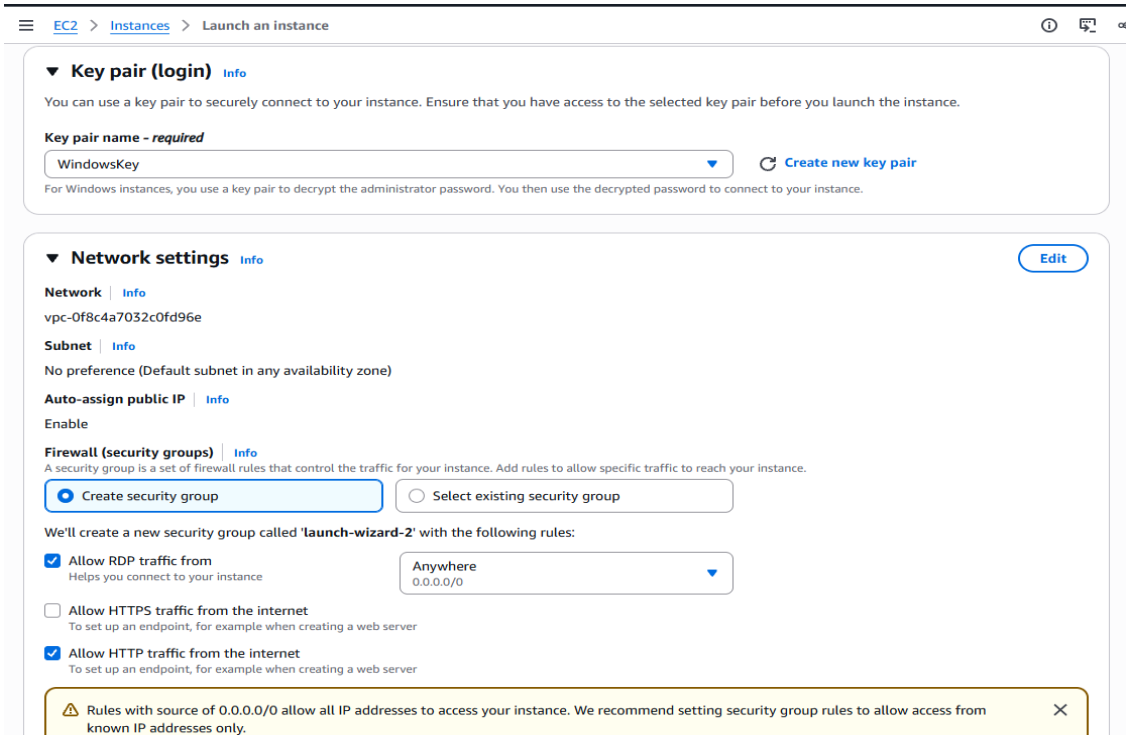
ii. Creating a Windows Virtual Machine

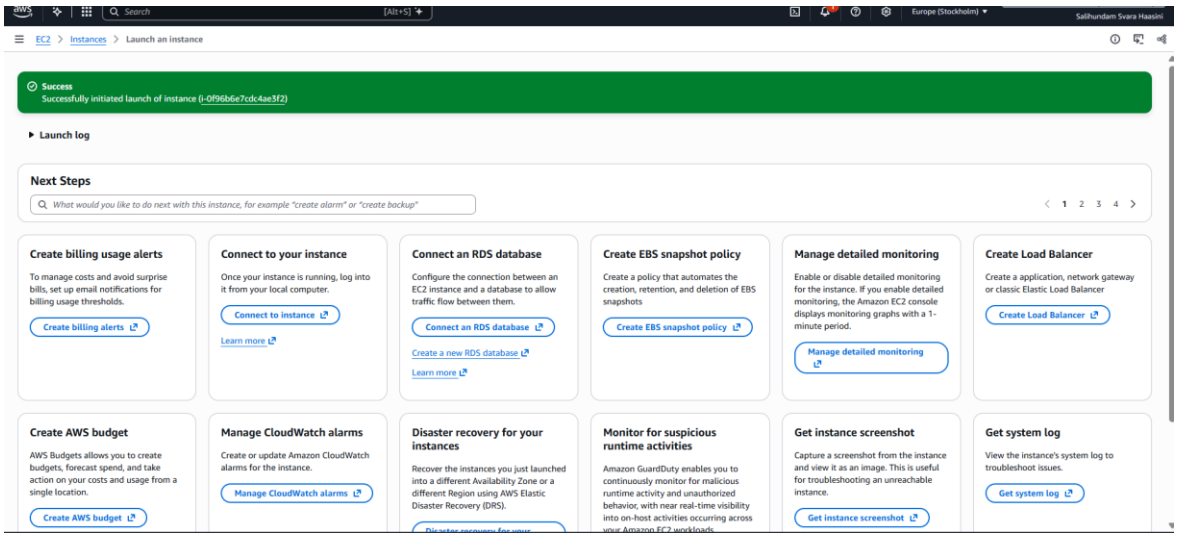


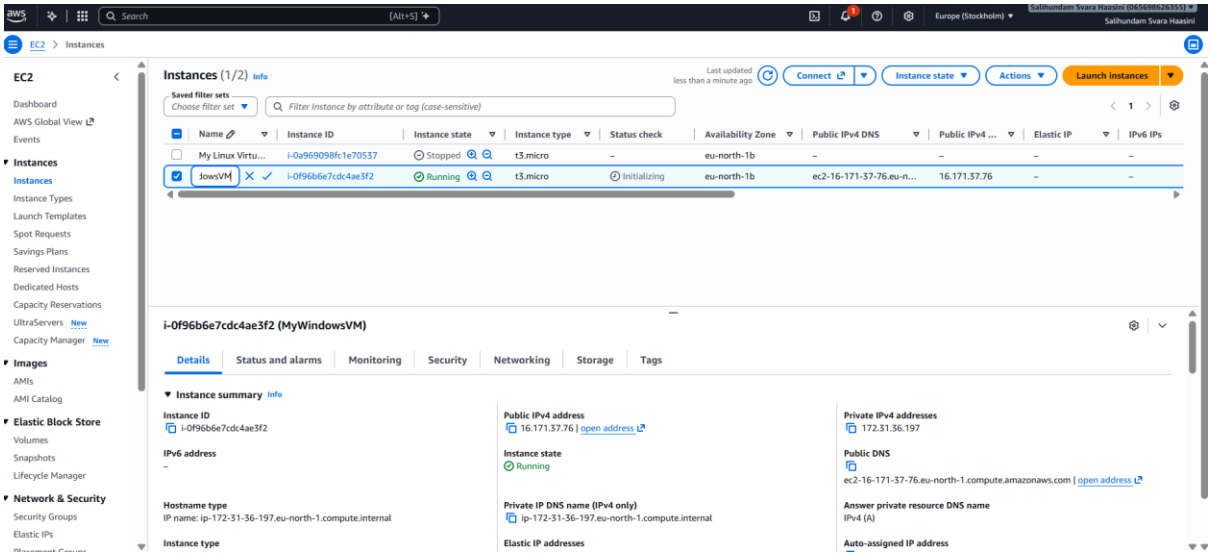


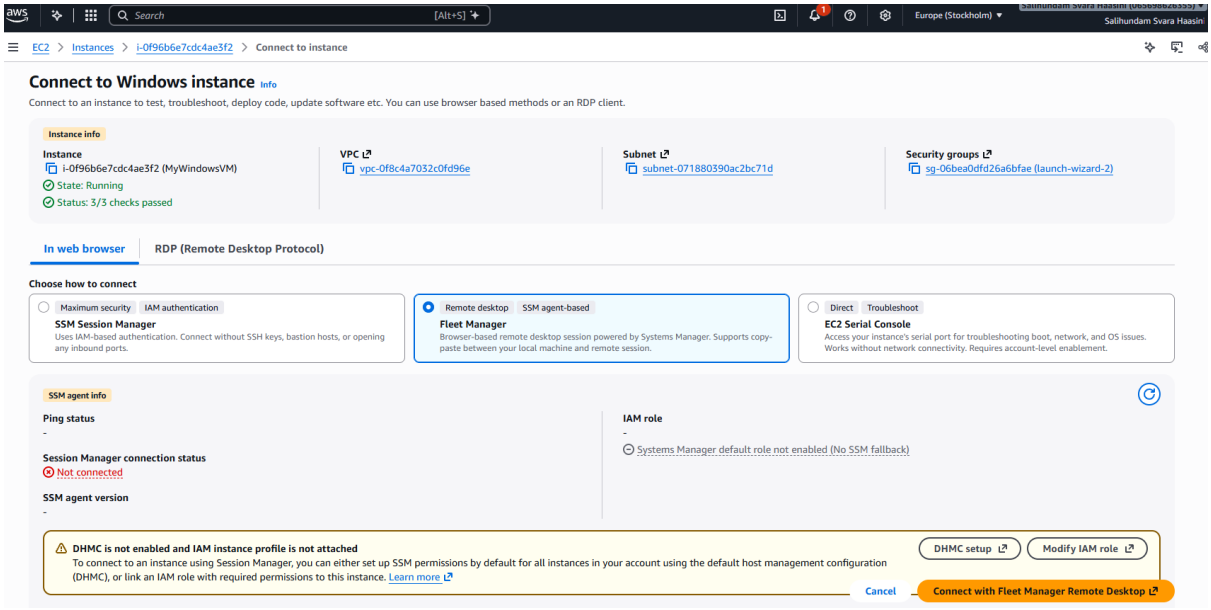

Launch an instance
Name and tags
MyWindowsVM
Application and OS Images (Amazon Machine Image)
Quick Start
Amazon Linux
macOS
Ubuntu
Windows
Red Hat
SUSE Linux
Debian
Browse more AMIs
Amazon Machine Image (AMI)
Microsoft Windows Server 2025 Base
Free tier eligible
Summary
Number of instances
Software Image (AMI)
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code

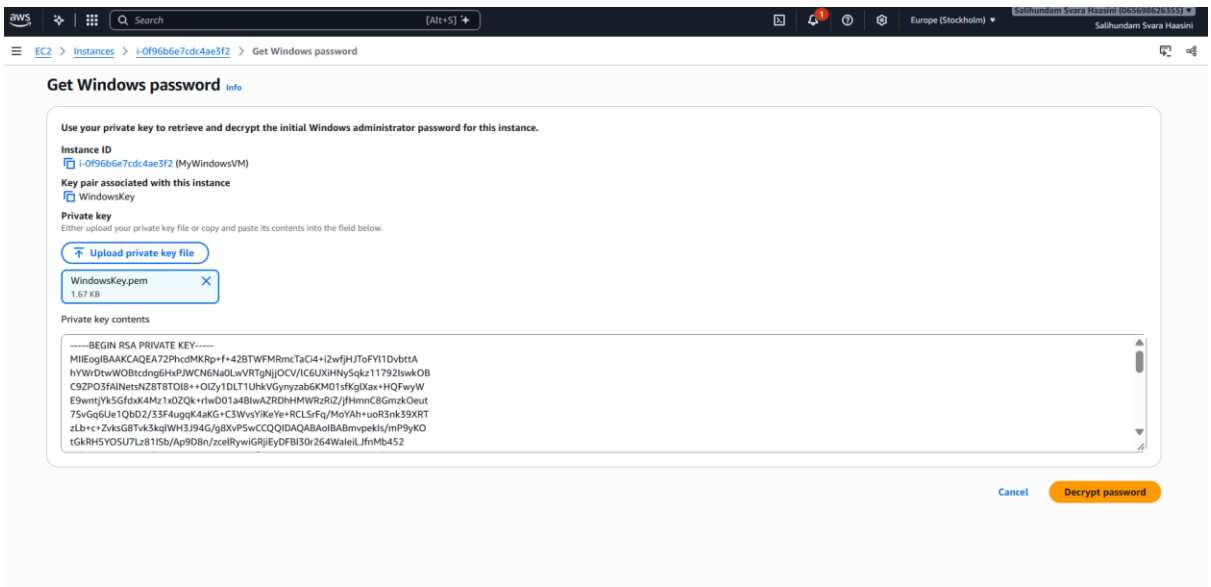

Create key pair
Key pair name
WindowsKey
Key pair type
RSA
ED25519
Private key file format
.pem
.ppk
When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance.
Cancel
Create key pair
Summary
Launch instance
Preview code

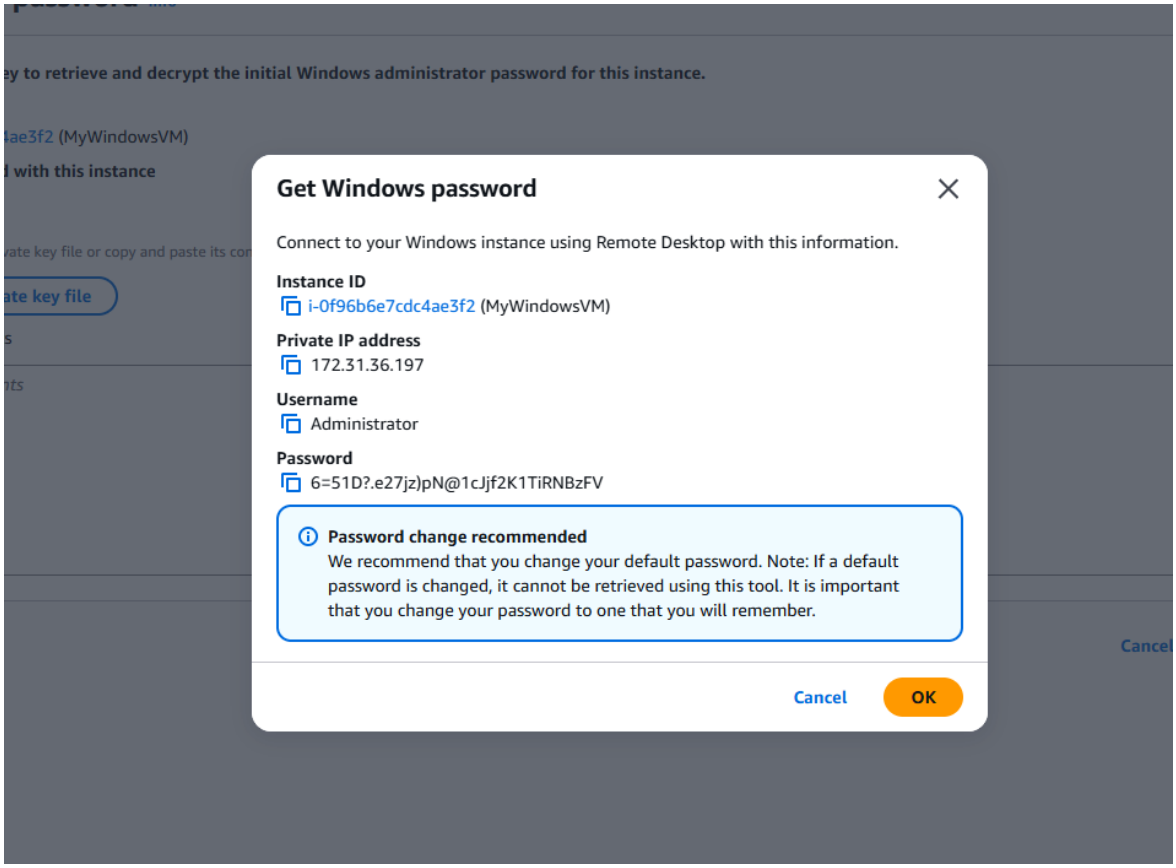

EC2 > Instances > Launch an instance
Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
WindowsKey
Create new key pair
For Windows instances, you use a key pair to decrypt the administrator password. You then use the decrypted password to connect to your instance.
Network settings Info
Edit
Network Info
vpc-0f8c4a7032c0fd96e
Subnet Info
No preference (Default subnet in any availability zone)
Auto-assign public IP Info
Enable
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
We'll create a new security group called 'launch-wizard-2' with the following rules:
Allow RDP traffic from
Helps you connect to your instance
Anywhere
0.0.0.0/0
Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server
Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server
Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

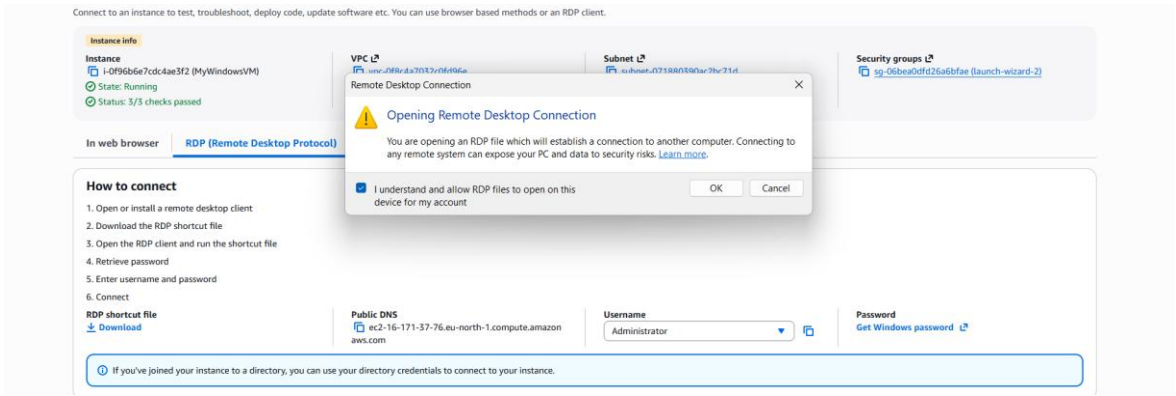

Success
Successfully initiated launch of instance (i-0f96b6e7cdc4ae3f2)
Launch log
Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"
Create billing usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.
Create billing alerts
Connect to your instance
Once your instance is running, log into it from your local computer.
Connect to instance
Learn more
Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
Connect an RDS database
Create a new RDS database
Learn more
Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots
Create EBS snapshot policy
Manage detailed monitoring
Enable or disable detailed monitoring for the instance. If you enable detailed monitoring, the Amazon EC2 console displays monitoring graphs with a 1-minute period.
Manage detailed monitoring
Create Load Balancer
Create a application, network gateway or classic Elastic Load Balancer
Create Load Balancer
Create AWS budget
AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage from a single location.
Create AWS budget
Manage CloudWatch alarms
Create or update Amazon CloudWatch alarms for the instance.
Manage CloudWatch alarms
Disaster recovery for your instances
Recover the instances you just launched into a different Availability Zone or a different Region using AWS Elastic Disaster Recovery (DRS).
Monitor for suspicious runtime activities
Amazon GuardDuty enables you to continuously monitor for malicious runtime activity and unauthorized behavior, with near real-time visibility into on-host activities occurring across
Get instance screenshot
Capture a screenshot from the instance and view it as an image. This is useful for troubleshooting an unreachable instance.
Get instance screenshot
Get system log
View the instance's system log to troubleshoot issues.
Get system log

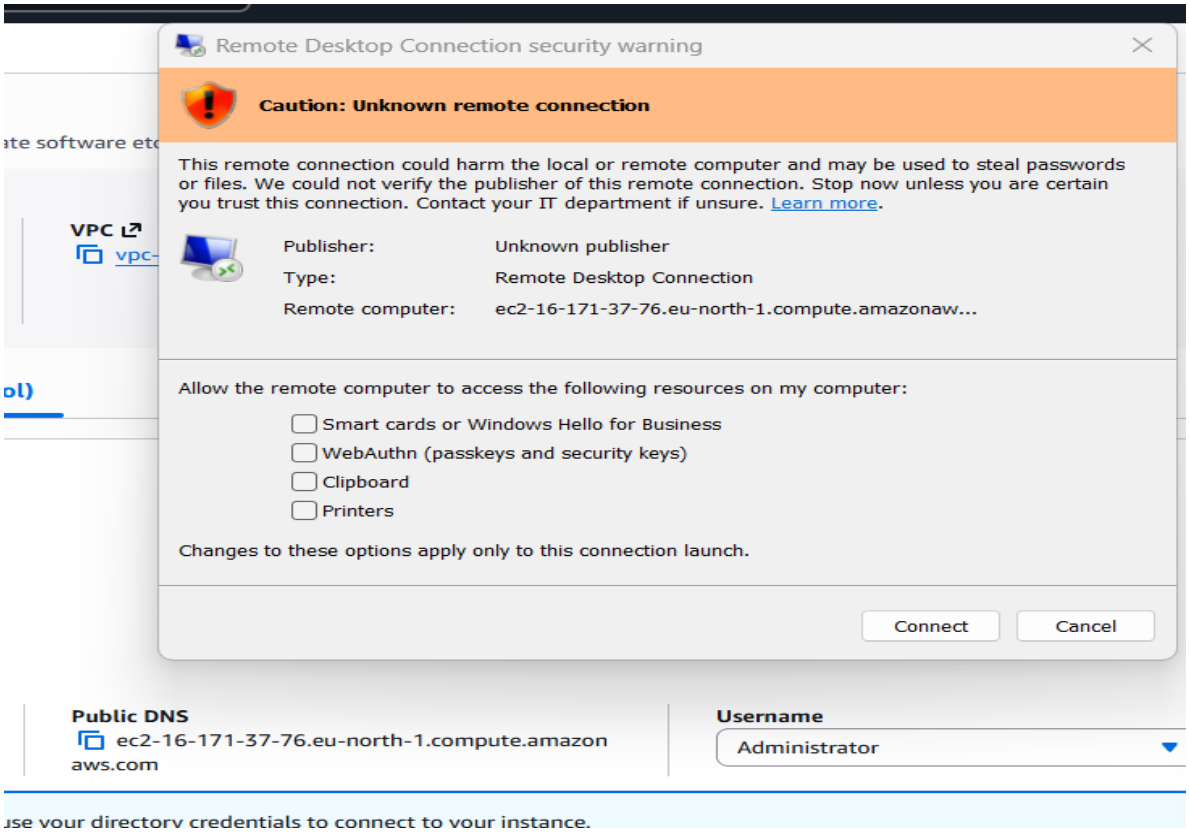

EC2 > Instances
Instances (1/2) Info
Connect
Instance state
Actions
Launch instances
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
Public IPv4 DNS
Public IPv4 ...
Elastic IP
IPv6 IPs
My Linux Virtu...
i-0a969098fc1e70537
Stopped
t3.micro
eu-north-1b
JowsVM
i-0f96b6e7cdc4ae3f2
Running
t3.micro
Initializing
eu-north-1b
ec2-16-171-37-76.eu-n...
16.171.37.76
i-0f96b6e7cdc4ae3f2 (MyWindowsVM)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-0f96b6e7cdc4ae3f2
Public IPv4 address
16.171.37.76 | open address
Private IPv4 addresses
172.31.36.197
IPv6 address
Instance state
Running
Public DNS
ec2-16-171-37-76.eu-north-1.compute.amazonaws.com | open address
Hostname type
IP name: ip-172-31-36-197.eu-north-1.compute.internal
Private IP DNS name (IPv4 only)
ip-172-31-36-197.eu-north-1.compute.internal
Answer private resource DNS name
IPv4 (A)
Instance type
Elastic IP addresses
Auto-assigned IP address

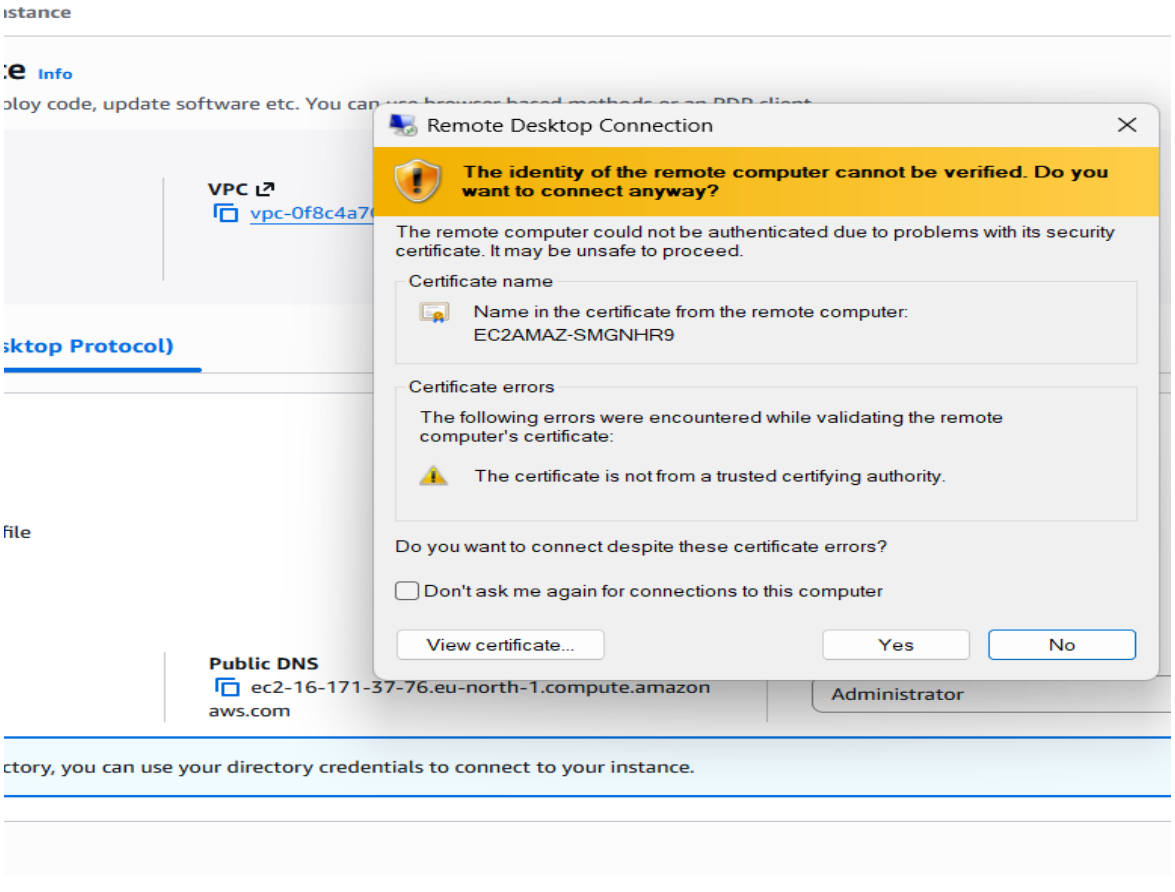

EC2 > Instances > i-0f96b6e7cdc4ae3f2 > Connect to instance
Connect to Windows instance Info
Connect to an instance to test, troubleshoot, deploy code, update software etc. You can use browser based methods or an RDP client.
Instance info
Instance
i-0f96b6e7cdc4ae3f2 (MyWindowsVM)
State: Running
Status: 3/3 checks passed
VPC
vpc-0f8c4a7032c0fd96e
Subnet
subnet-071880390ac2bc71d
Security groups
sg-06bea0dfd26a6bfae (launch-wizard-2)
In web browser
RDP (Remote Desktop Protocol)
Choose how to connect
Maximum security
IAM authentication
SSM Session Manager
Uses IAM-based authentication. Connect without SSH keys, bastion hosts, or opening any inbound ports.
Remote desktop
SSM agent-based
Fleet Manager
Browser-based remote desktop session powered by Systems Manager. Supports copy-paste between your local machine and remote session.
Direct
Troubleshoot
EC2 Serial Console
Access your instance's serial port for troubleshooting boot, network, and OS issues. Works without network connectivity. Requires account-level enablement.
SSM agent info
Ping status
-
Session Manager connection status
Not connected
SSM agent version
-
IAM role
-
Systems Manager default role not enabled (No SSM fallback)
DHMC is not enabled and IAM instance profile is not attached
To connect to an instance using Session Manager, you can either set up SSM permissions by default for all instances in your account using the default host management configuration (DHMC), or link an IAM role with required permissions to this instance. Learn more
DHMC setup
Modify IAM role
Cancel
Connect with Fleet Manager Remote Desktop

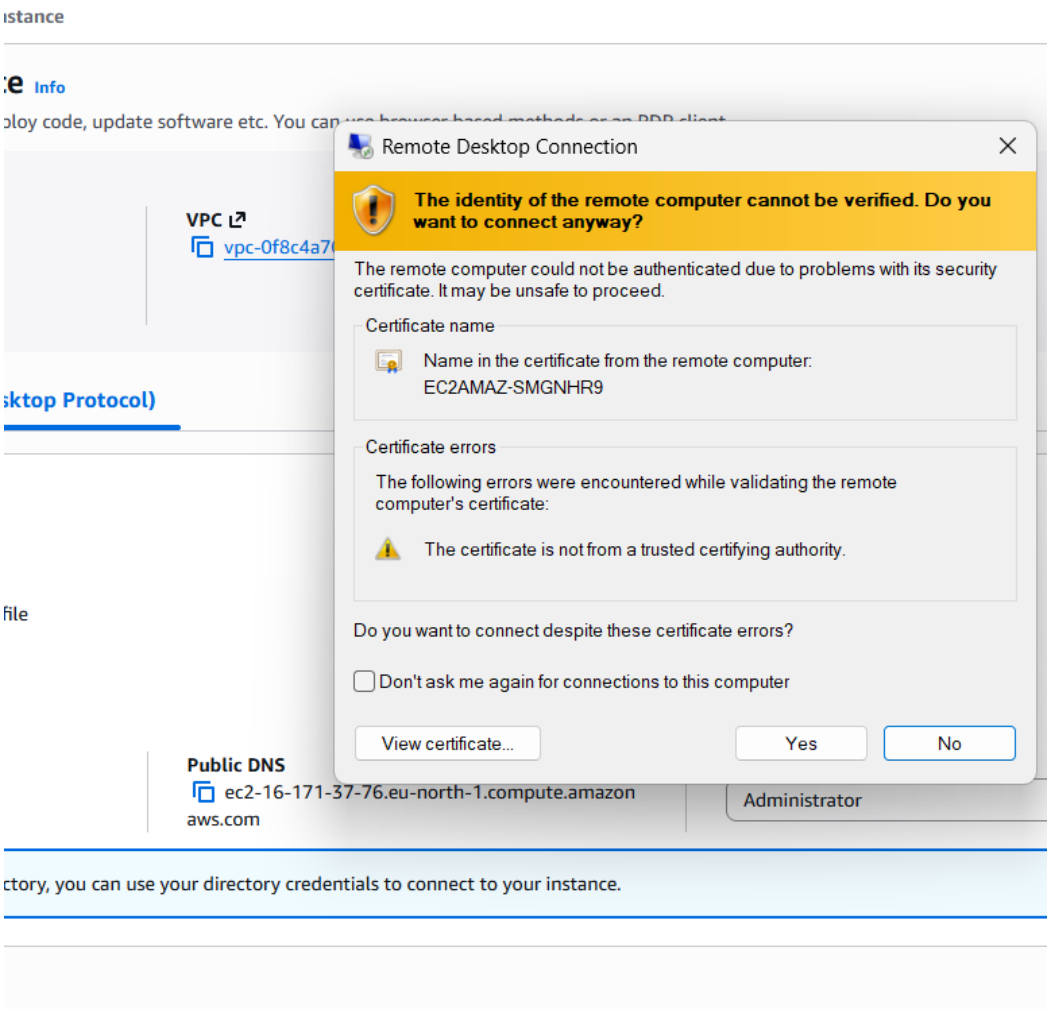

EC2 > Instances > i-0f96b6e7cdc4ae3f2 > Get Windows password
Get Windows password Info
Use your private key to retrieve and decrypt the initial Windows administrator password for this instance.
Instance ID
i-0f96b6e7cdc4ae3f2 (MyWindowsVM)
Key pair associated with this instance
WindowsKey
Private key
Either upload your private key file or copy and paste its contents into the field below.
Upload private key file
WindowsKey.pem
1.67 KB
Private key contents
-----BEGIN RSA PRIVATE KEY-----
MIIEogIBAAKCAQEA72PhcdMKRp+f+42BTWFMRmcTaCi4+I2wfjHJToFYl1DvbttA
hYWrDtwWOBtcdng6HxPJWCN6Na0LwVRTgNjjOCV/IC6UXiHNySqkz11792IswkOB
C9ZPO3fAlNetsNZ8T8TOI8++OIZy1DLT1UhkVGynyzab6KM01sfKglXax+HQFwyW
E9wntjYk5GfdxK4Mz1x0ZQk+rlwD01a4BIwAZRDhHMWRzRiZ/jfHmnC8GmzkOeut
7SvGq6Ue1QbD2/33F4ugqK4aKG+C3WvsYiKeYe+RCLSrFq/MoYAh+uoR3nk39XRT
zLb+c+ZvksG8Tvk3kqlWH3J94G/g8XvP5wCCQQIDAQABAoIBABmvpekIs/mP9yKO
tGkRH5YOSU7Lz81ISb/Ap9D8n/zcelRywiGRjiEyDFBI30r264WaIeiLJfnMb452
Cancel
Decrypt password

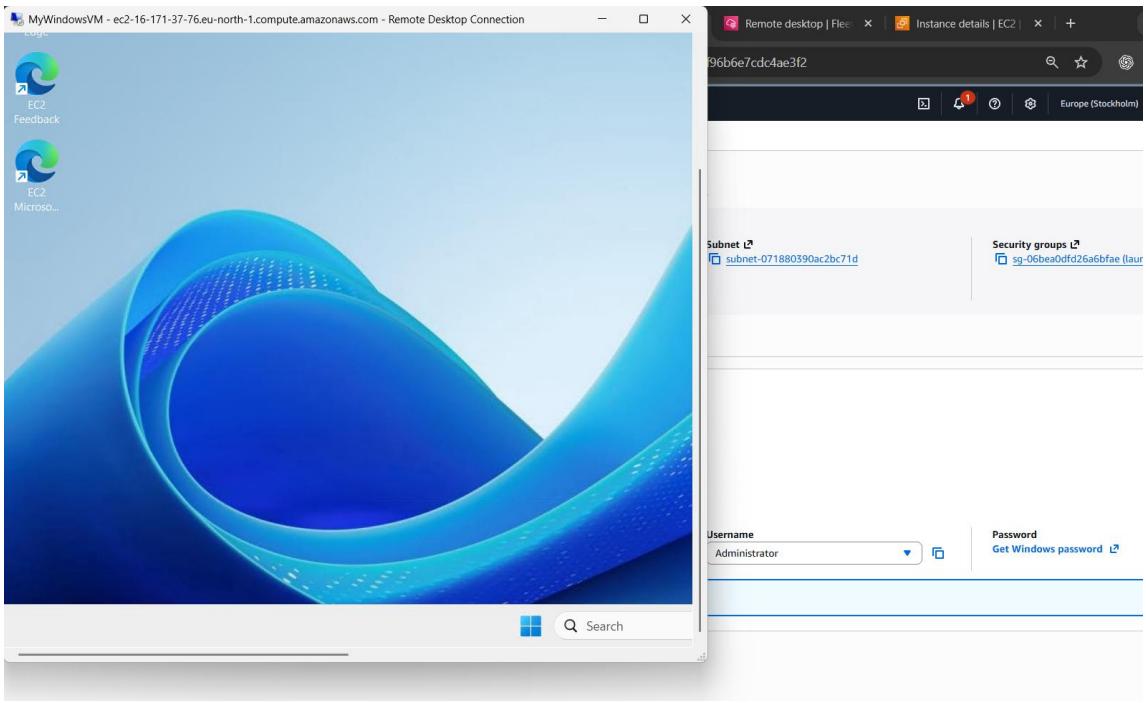

ey to retrieve and decrypt the initial Windows administrator password for this instance.
4ae3f2 (MyWindowsVM)
d with this instance
vate key file or copy and paste its con
ate key file
Get Windows password
Connect to your Windows instance using Remote Desktop with this information.
Instance ID
i-0f96b6e7cdc4ae3f2 (MyWindowsVM)
Private IP address
172.31.36.197
Username
Administrator
Password
6=51D?.e27jz)pN@1cJjf2K1TiRNBzFV
Password change recommended
We recommend that you change your default password. Note: If a default password is changed, it cannot be retrieved using this tool. It is important that you change your password to one that you will remember.
Cancel
OK


Connect to an instance to test, troubleshoot, deploy code, update software etc. You can use browser based methods or an RDP client.
Instance info
Instance
i-0f96b6e7cdc4ae3f2 (MyWindowsVM)
State: Running
Status: 3/3 checks passed
VPC
Subnet
Security groups
sg-06bea0dfd26a6bfae (launch-wizard-2)
Remote Desktop Connection
Opening Remote Desktop Connection
You are opening an RDP file which will establish a connection to another computer. Connecting to any remote system can expose your PC and data to security risks. Learn more.
I understand and allow RDP files to open on this device for my account
OK
Cancel
In web browser
RDP (Remote Desktop Protocol)
How to connect
1. Open or install a remote desktop client
2. Download the RDP shortcut file
3. Open the RDP client and run the shortcut file
4. Retrieve password
5. Enter username and password
6. Connect
RDP shortcut file
Download
Public DNS
ec2-16-171-37-76.eu-north-1.compute.amazonaws.com
Username
Administrator
Password
Get Windows password
If you've joined your instance to a directory, you can use your directory credentials to connect to your instance.

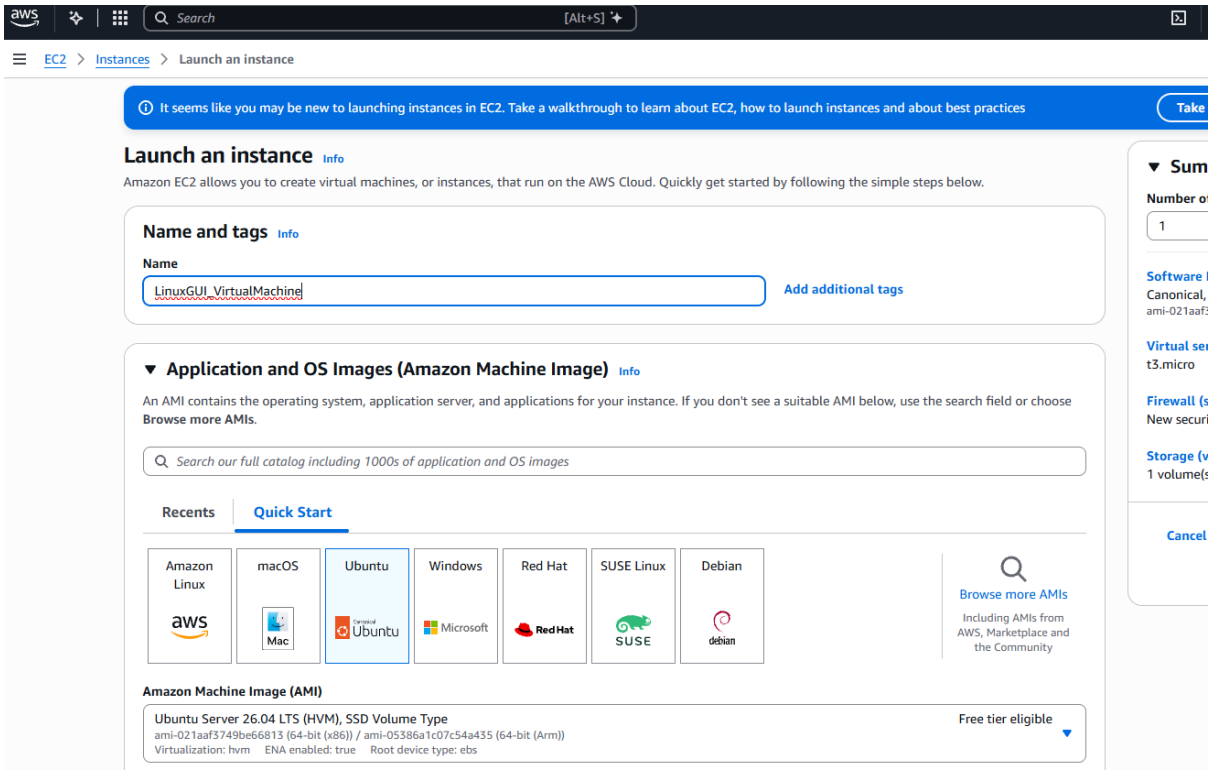

Remote Desktop Connection security warning
Caution: Unknown remote connection
This remote connection could harm the local or remote computer and may be used to steal passwords or files. We could not verify the publisher of this remote connection. Stop now unless you are certain you trust this connection. Contact your IT department if unsure. Learn more.
Publisher: Unknown publisher
Type: Remote Desktop Connection
Remote computer: ec2-16-171-37-76.eu-north-1.compute.amazonaw...
Allow the remote computer to access the following resources on my computer:
Smart cards or Windows Hello for Business
WebAuthn (passkeys and security keys)
Clipboard
Printers
Changes to these options apply only to this connection launch.
Connect
Cancel
VPC
Public DNS
ec2-16-171-37-76.eu-north-1.compute.amazonaws.com
Username
Administrator

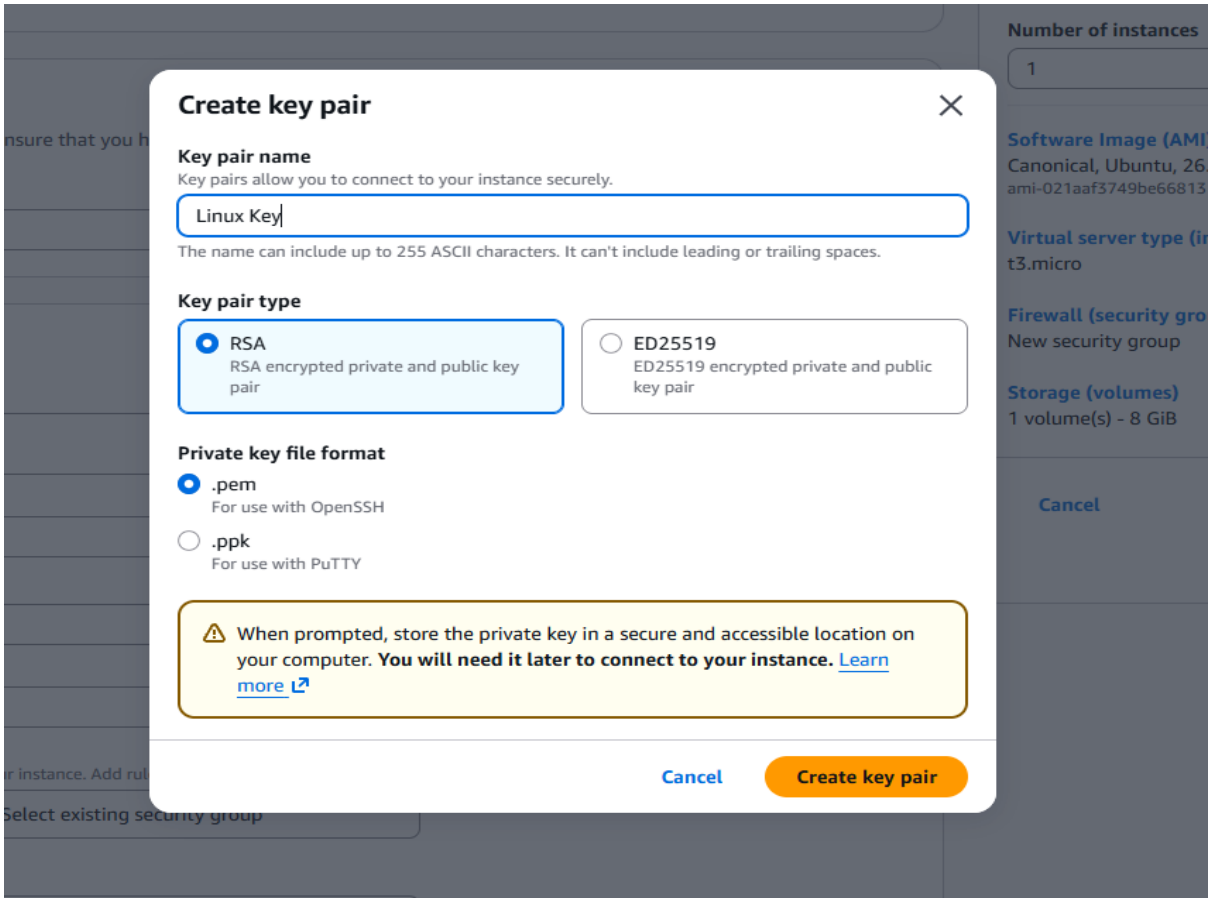

Info
Remote Desktop Connection
The identity of the remote computer cannot be verified. Do you want to connect anyway?
The remote computer could not be authenticated due to problems with its security certificate. It may be unsafe to proceed.
Certificate name
Name in the certificate from the remote computer:
EC2AMAZ-SMGNHR9
Certificate errors
The following errors were encountered while validating the remote computer's certificate:
The certificate is not from a trusted certifying authority.
Do you want to connect despite these certificate errors?
Don't ask me again for connections to this computer
View certificate...
Yes
No
VPC
Public DNS
ec2-16-171-37-76.eu-north-1.compute.amazonaws.com
Administrator

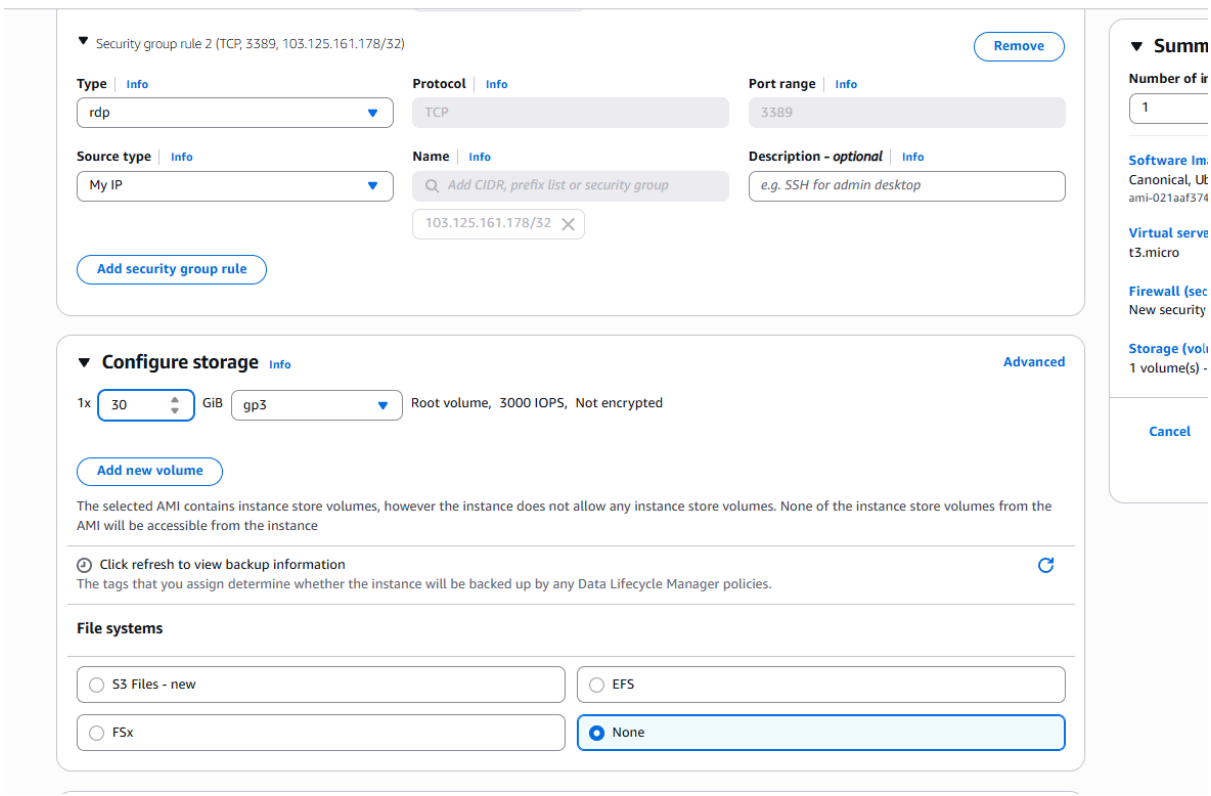

Info
VPC
vpc-0f8c4a7
Remote Desktop Connection
The identity of the remote computer cannot be verified. Do you want to connect anyway?
The remote computer could not be authenticated due to problems with its security certificate. It may be unsafe to proceed.
Certificate name
Name in the certificate from the remote computer:
EC2AMAZ-SMGNHR9
Certificate errors
The following errors were encountered while validating the remote computer's certificate:
The certificate is not from a trusted certifying authority.
Do you want to connect despite these certificate errors?
Don't ask me again for connections to this computer
View certificate...
Yes
No
Public DNS
ec2-16-171-37-76.eu-north-1.compute.amazonaws.com
Administrator
ctory, you can use your directory credentials to connect to your instance.

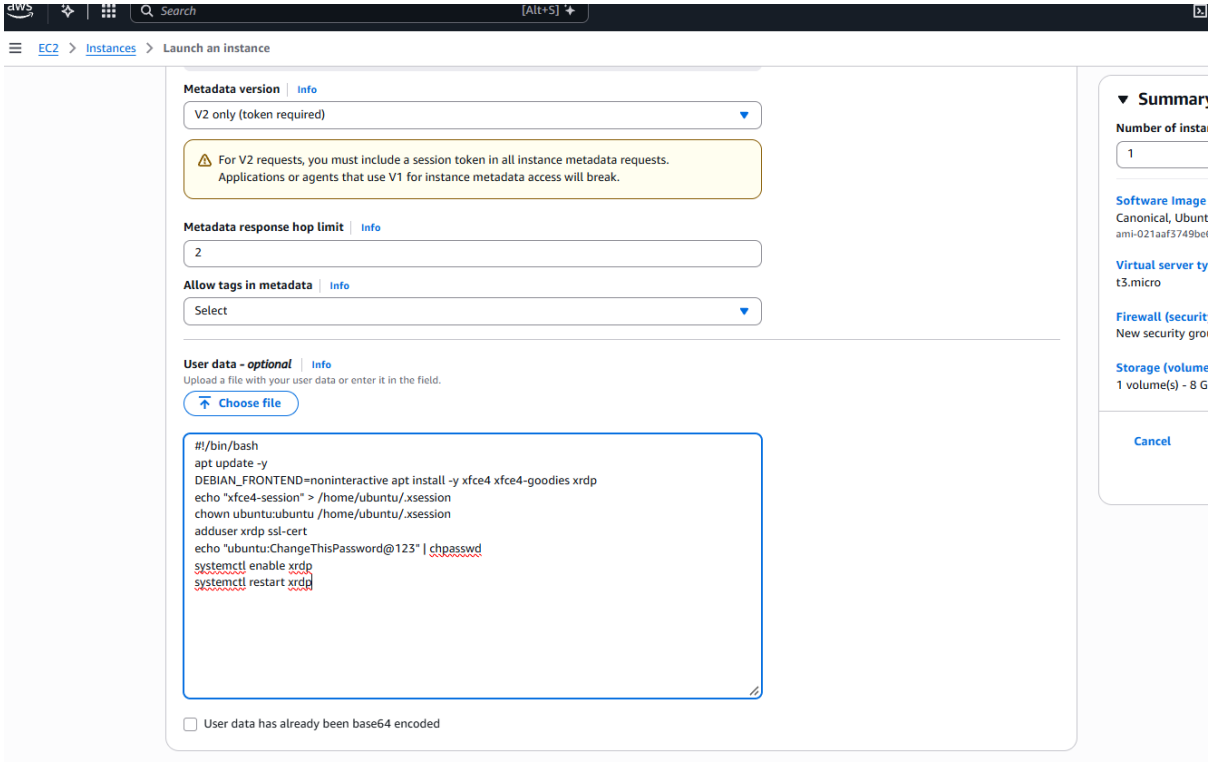

MyWindowsVM - ec2-16-171-37-76.eu-north-1.compute.amazonaws.com - Remote Desktop Connection
EC2 Feedback
EC2 Microso...
Search
Remote desktop | Flee
Instance details | EC2
Europe (Stockholm)
Subnet
subnet-071880390ac2bc71d
Security groups
Username
Administrator
Password
Get Windows password

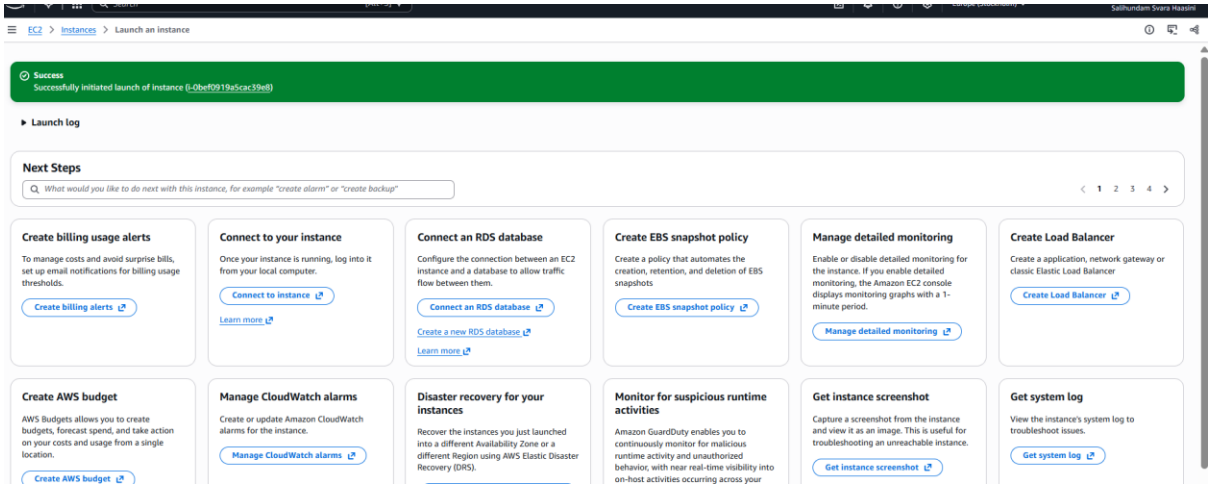
iii. Creating a LinuxGUI Virtual Machine

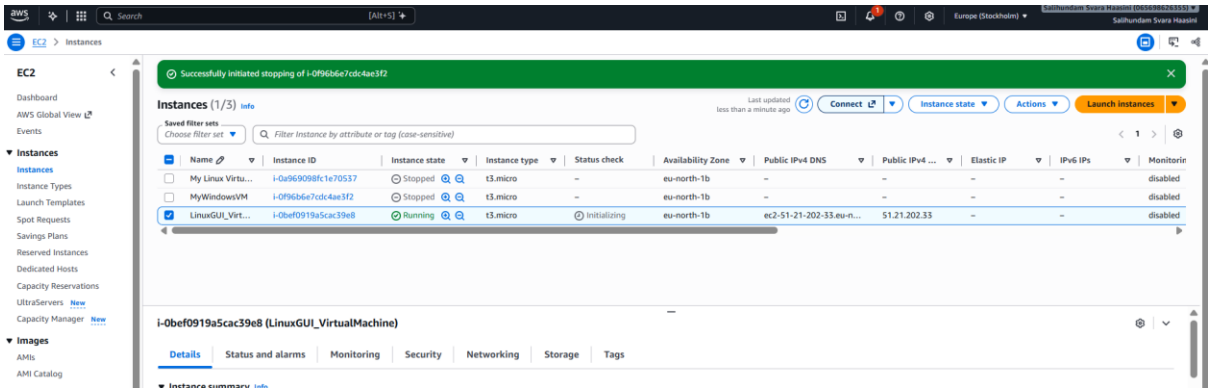


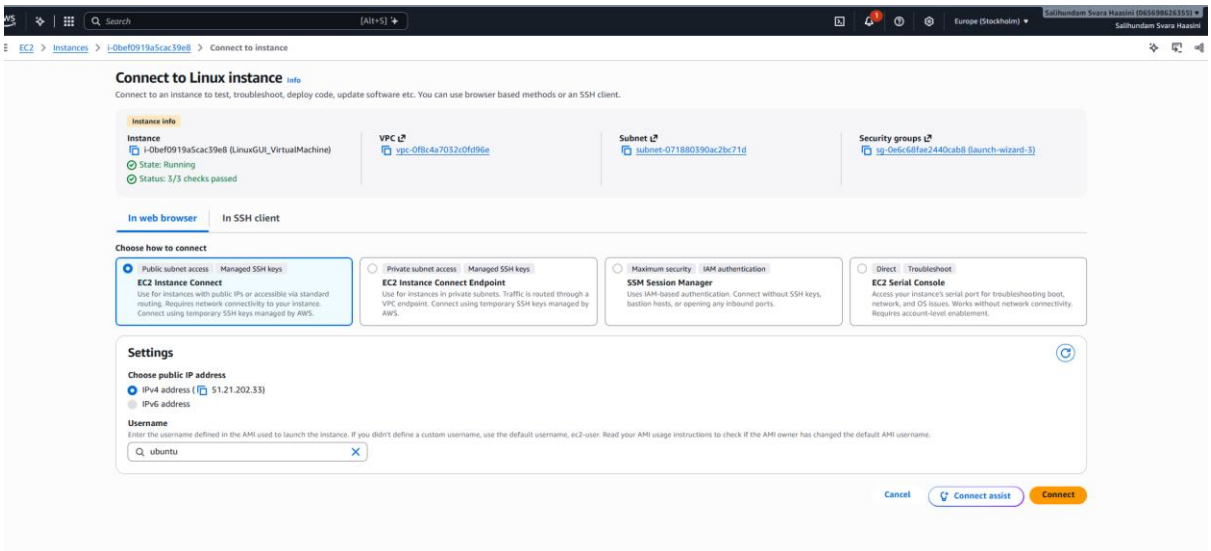


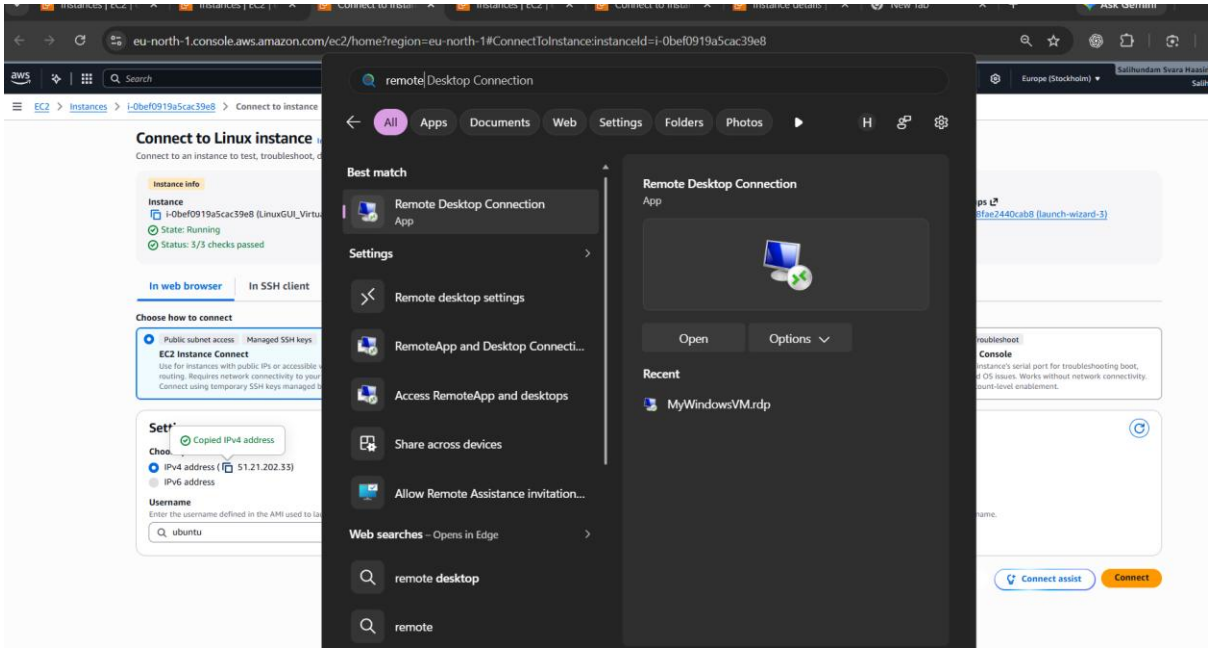

Security group rule 2 (TCP, 3389, 103.125.161.178/32)
Remove
Type | Info
rdp
Protocol | Info
TCP
Port range | Info
3389
Source type | Info
My IP
Name | Info
Add CIDR, prefix list or security group
103.125.161.178/32
Description - optional | Info
e.g. SSH for admin desktop
Add security group rule
Configure storage Info
Advanced
1x
30
GiB
gp3
Root volume, 3000 IOPS, Not encrypted
Add new volume
The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance
Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.
File systems
S3 Files - new
EFS
FSx
None
Summ
Number of ir
1
Software Ima
Canonical, Ub
ami-021aaf374
Virtual serve
t3.micro
Firewall (sec
New security
Storage (volu
1 volume(s) -
Cancel

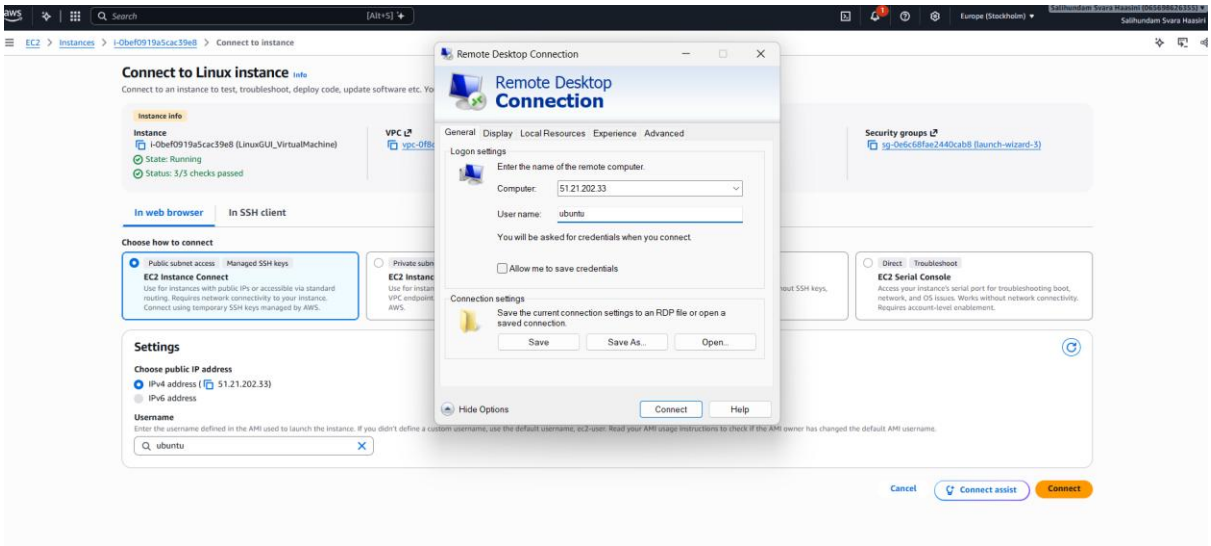

Search
[Alt+S]
EC2 > Instances > Launch an instance
Metadata version | Info
V2 only (token required)
For V2 requests, you must include a session token in all instance metadata requests.
Applications or agents that use V1 for instance metadata access will break.
Metadata response hop limit | Info
2
Allow tags in metadata | Info
Select
User data - optional | Info
Upload a file with your user data or enter it in the field.
Choose file
#!/bin/bash
apt update -y
DEBIAN_FRONTEND=noninteractive apt install -y xfce4 xfce4-goodies xrdp
echo "xfce4-session" > /home/ubuntu/.xsession
chown ubuntu:ubuntu /home/ubuntu/.xsession
adduser xrdp ssl-cert
echo "ubuntu:ChangeThisPassword@123" | chpasswd
systemctl enable xrdp
systemctl restart xrdp
User data has already been base64 encoded
Summary
Number of insta
1
Software Image
Canonical, Ubunt
ami-021aaf3749be
Virtual server ty
t3.micro
Firewall (securit
New security gro
Storage (volume
1 volume(s) - 8 G
Cancel

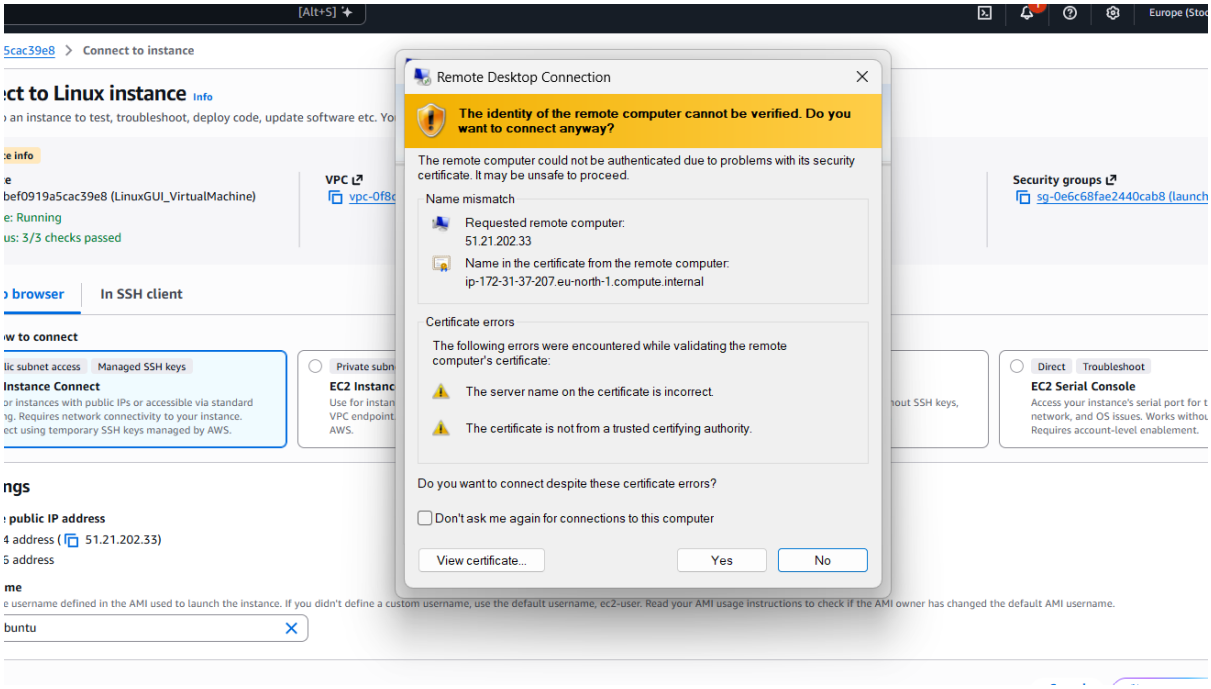

EC2 > Instances > Launch an instance
Success
Successfully initiated launch of instance (i-0bef0919a5cac39e8)
Launch log
Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"
Create billing usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.
Create billing alerts
Connect to your instance
Once your instance is running, log into it from your local computer.
Connect to instance
Learn more
Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
Connect an RDS database
Create a new RDS database
Learn more
Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots
Create EBS snapshot policy
Manage detailed monitoring
Enable or disable detailed monitoring for the instance. If you enable detailed monitoring, the Amazon EC2 console displays monitoring graphs with a 1-minute period.
Manage detailed monitoring
Create Load Balancer
Create a application, network gateway or classic Elastic Load Balancer
Create Load Balancer
Create AWS budget
AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage from a single location.
Create AWS budget
Manage CloudWatch alarms
Create or update Amazon CloudWatch alarms for the instance.
Manage CloudWatch alarms
Disaster recovery for your instances
Recover the instances you just launched into a different Availability Zone or a different Region using AWS Elastic Disaster Recovery (DRS).
Monitor for suspicious runtime activities
Amazon GuardDuty enables you to continuously monitor for malicious runtime activity and unauthorized behavior, with near real-time visibility into on-host activities occurring across your
Get instance screenshot
Capture a screenshot from the instance and view it as an image. This is useful for troubleshooting an unreachable instance.
Get instance screenshot
Get system log
View the instance's system log to troubleshoot issues.
Get system log

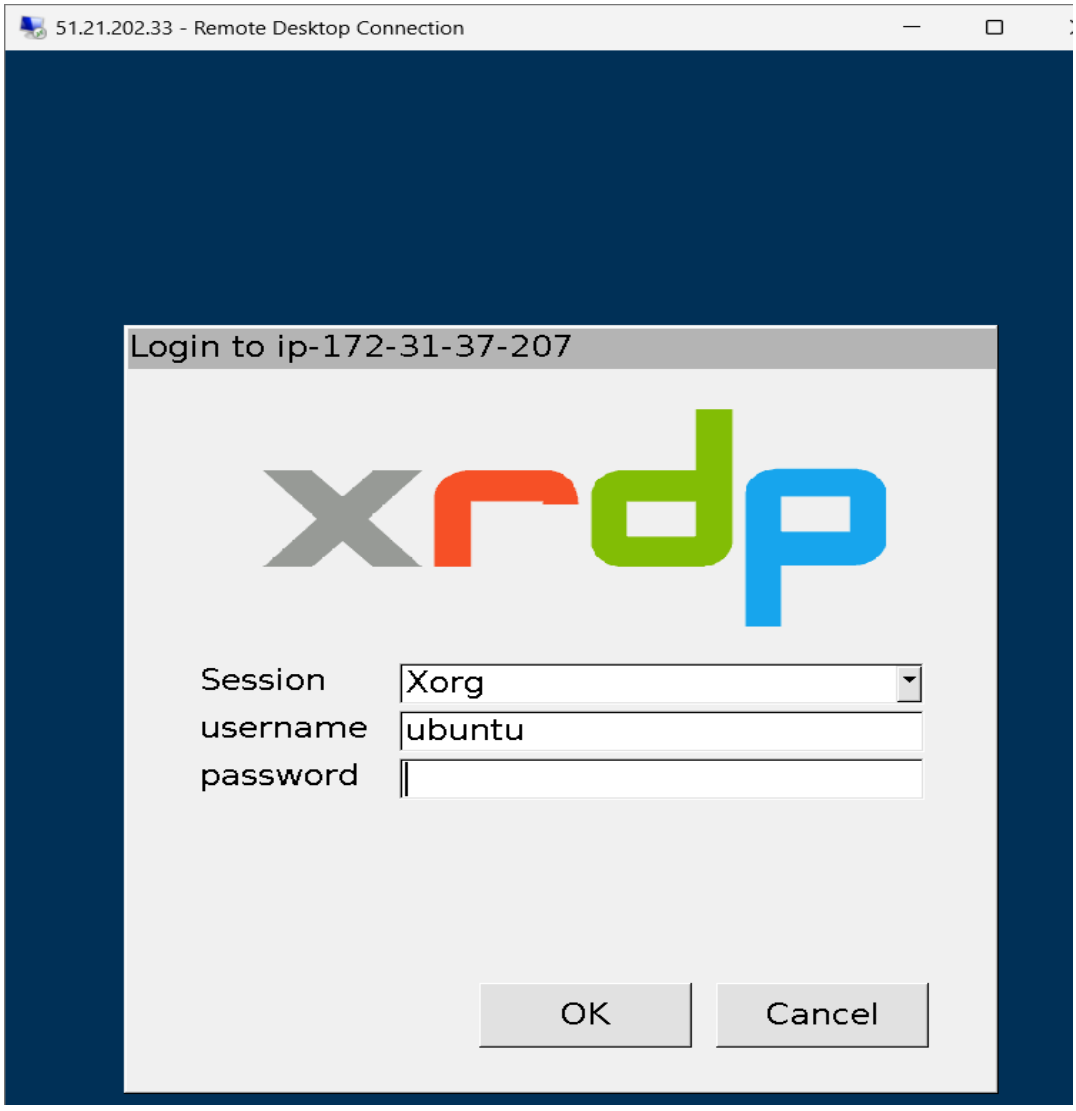

EC2 > Instances
Successfully initiated stopping of i-0f96b6e7cdc4ae3f2
Instances (1/3) Info
Connect
Instance state
Actions
Launch instances
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
Public IPv4 DNS
Public IPv4 ...
Elastic IP
IPv6 IPs
Monitorin
My Linux Virtu...
i-0a969098fc1e70537
Stopped
t3.micro
eu-north-1b
disabled
MyWindowsVM
i-0f96b6e7cdc4ae3f2
Stopped
t3.micro
eu-north-1b
disabled
LinuxGUI_Virt...
i-0bef0919a5cac39e8
Running
t3.micro
Initializing
eu-north-1b
ec2-51-21-202-33.eu-n...
51.21.202.33
disabled
i-0bef0919a5cac39e8 (LinuxGUI_VirtualMachine)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags

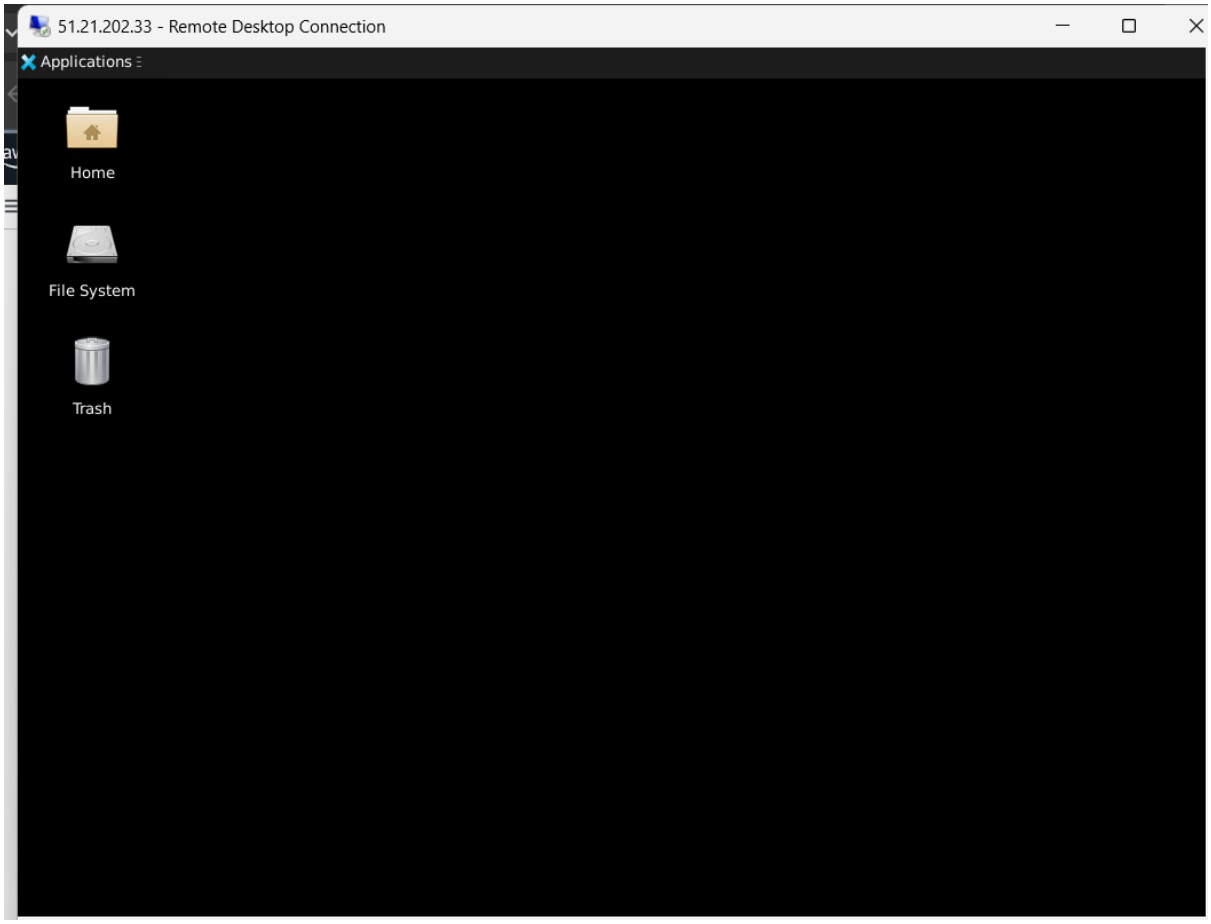

EC2 > Instances > i-0bef0919a5cac39e8 > Connect to instance
Connect to Linux instance Info
Connect to an instance to test, troubleshoot, deploy code, update software etc. You can use browser based methods or an SSH client.
Instance info
Instance
i-0bef0919a5cac39e8 (LinuxGUI_VirtualMachine)
State: Running
Status: 3/3 checks passed
VPC
vpc-0f8c4a7032c0fd96e
Subnet
subnet-071880390ac2bc71d
Security groups
sg-0e6c68fae2440cab8 (launch-wizard-3)
In web browser
In SSH client
Choose how to connect
EC2 Instance Connect
Use for instances with public IPs or accessible via standard routing. Requires network connectivity to your instance. Connect using temporary SSH keys managed by AWS.
EC2 Instance Connect Endpoint
Use for instances in private subnets. Traffic is routed through a VPC endpoint. Connect using temporary SSH keys managed by AWS.
SSM Session Manager
Uses IAM-based authentication. Connect without SSH keys, bastion hosts, or opening any inbound ports.
EC2 Serial Console
Access your instance's serial port for troubleshooting boot, network, and OS issues. Works without network connectivity. Requires account-level enablement.
Settings
Choose public IP address
IPv4 address (51.21.202.33)
IPv6 address
Username
ubuntu
Cancel
Connect assist
Connect

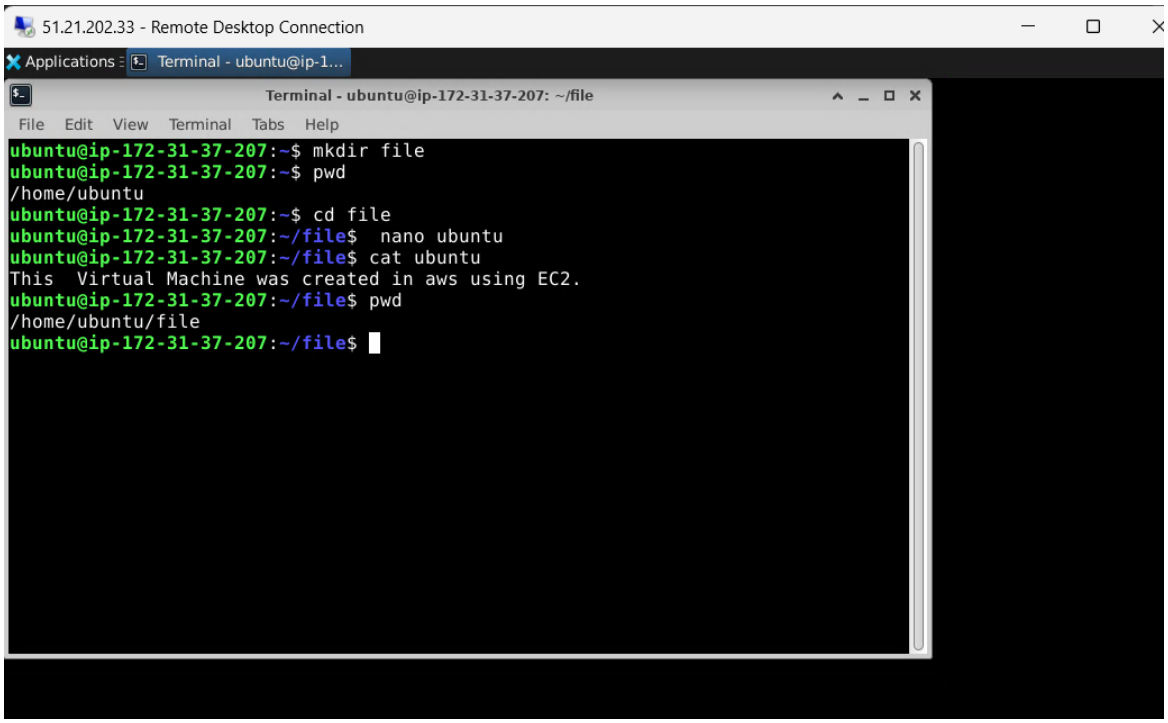

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#ConnectToInstance:instanceId=i-0bef0919a5cac39e8
remote Desktop Connection
All
Apps
Documents
Web
Settings
Folders
Photos
Best match
Remote Desktop Connection
App
Settings
Remote desktop settings
RemoteApp and Desktop Connecti...
Access RemoteApp and desktops
Share across devices
Allow Remote Assistance invitation...
Web searches – Opens in Edge
remote desktop
remote
Remote Desktop Connection
App
Open
Options
Recent
MyWindowsVM.rdp
Connect to Linux instance
In web browser
In SSH client
Choose how to connect
EC2 Instance Connect
Copied IPv4 address
IPv4 address (51.21.202.33)
IPv6 address
Username
ubuntu
Connect assist
Connect

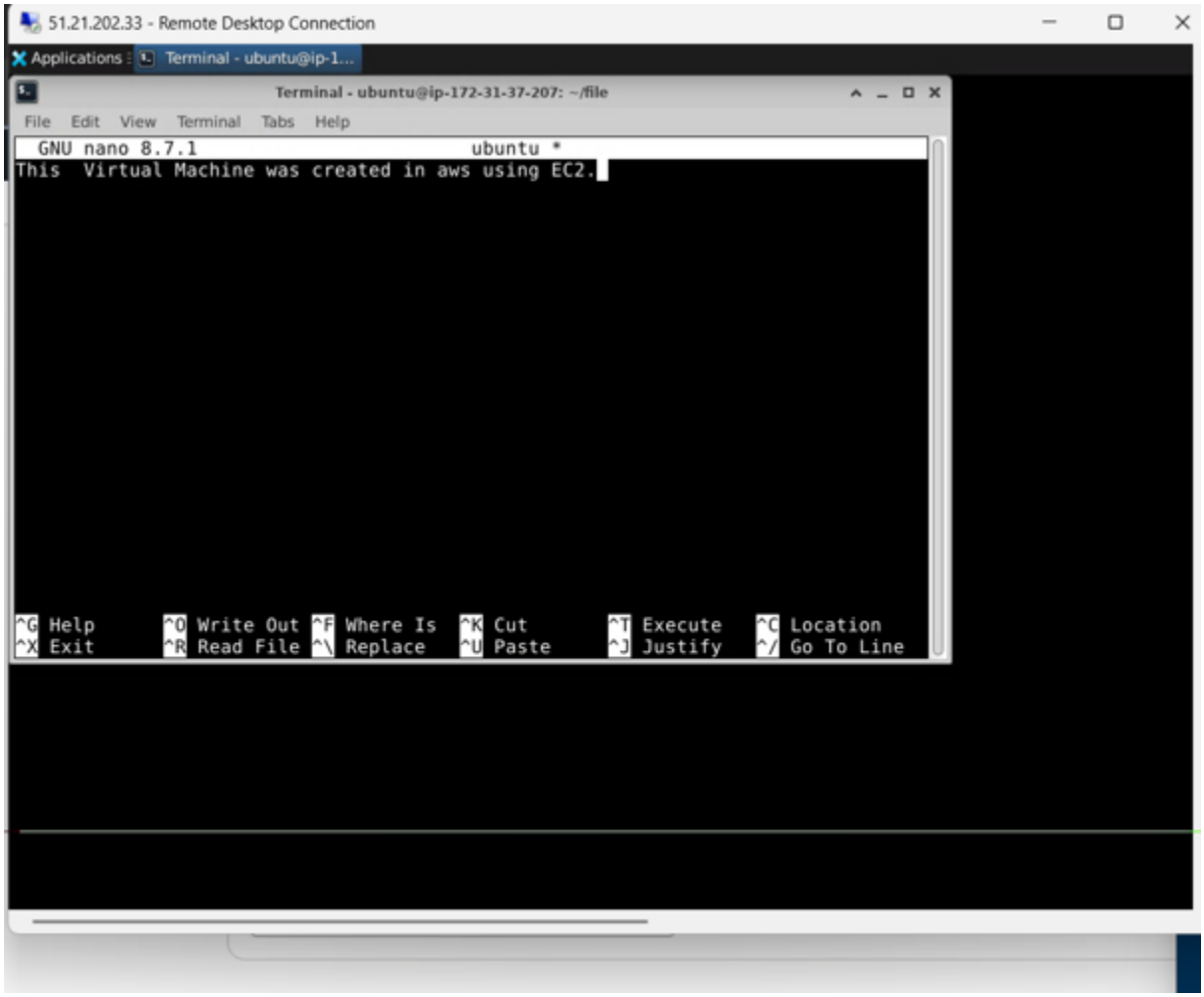

Connect to Linux instance
Remote Desktop Connection
Remote Desktop Connection
General
Display
Local Resources
Experience
Advanced
Logon settings
Enter the name of the remote computer.
Computer:
51.21.202.33
User name:
ubuntu
You will be asked for credentials when you connect.
Allow me to save credentials
Connection settings
Save the current connection settings to an RDP file or open a saved connection.
Save
Save As...
Open...
Hide Options
Connect
Help
In web browser
In SSH client
Choose how to connect
EC2 Instance Connect
EC2 Serial Console
Settings
Choose public IP address
IPv4 address (51.21.202.33)
IPv6 address
Username
ubuntu
Cancel
Connect assist
Connect


Remote Desktop Connection
The identity of the remote computer cannot be verified. Do you want to connect anyway?
The remote computer could not be authenticated due to problems with its security certificate. It may be unsafe to proceed.
Name mismatch
Requested remote computer:
51.21.202.33
Name in the certificate from the remote computer:
ip-172-31-37-207.eu-north-1.compute.internal
Certificate errors
The following errors were encountered while validating the remote computer's certificate:
The server name on the certificate is incorrect.
The certificate is not from a trusted certifying authority.
Do you want to connect despite these certificate errors?
Don't ask me again for connections to this computer
View certificate...
Yes
No
Connect to instance
Connect to Linux instance
In SSH client
EC2 Instance Connect
EC2 Serial Console
Security groups


51.21.202.33 - Remote Desktop Connection
Login to ip-172-31-37-207
xrdp
Session
Xorg
username
ubuntu
password
OK
Cancel

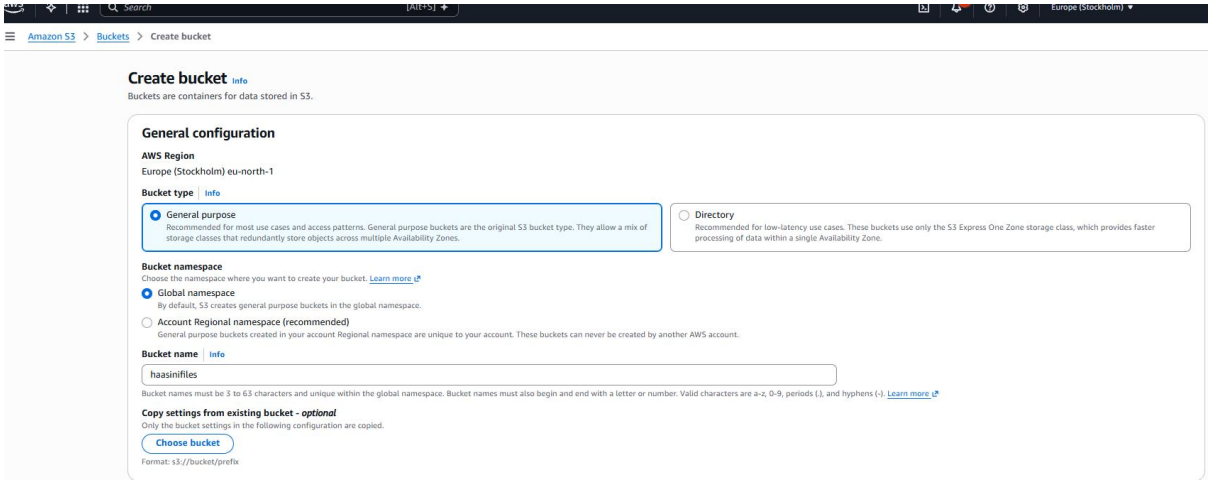

51.21.202.33 - Remote Desktop Connection
Applications
Home
File System
Trash

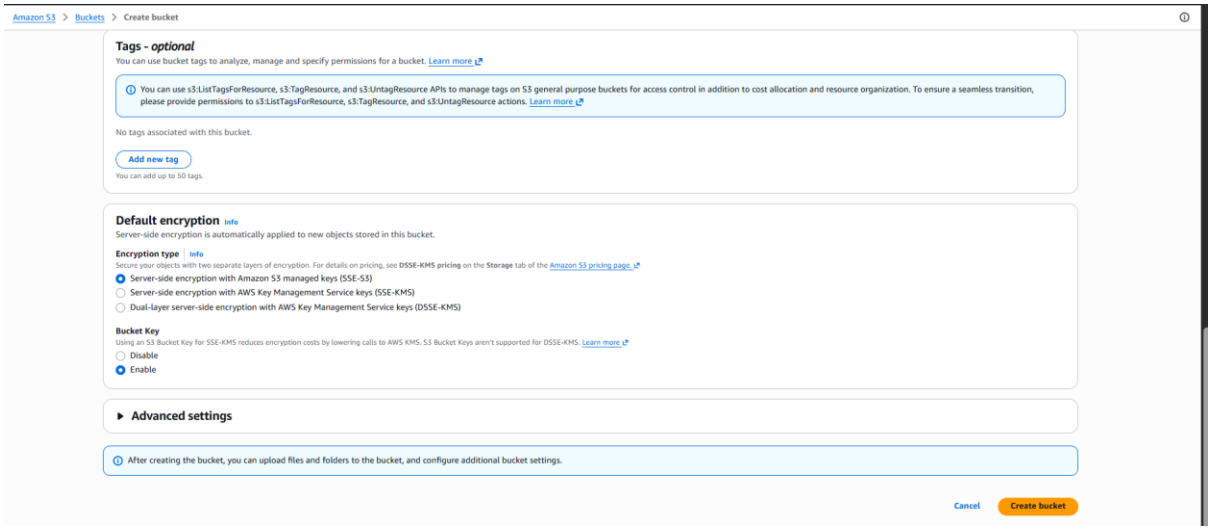

51.21.202.33 - Remote Desktop Connection
Applications
Terminal - ubuntu@ip-1...
Terminal - ubuntu@ip-172-31-37-207: ~/file
File Edit View Terminal Tabs Help
ubuntu@ip-172-31-37-207:~$ mkdir file
ubuntu@ip-172-31-37-207:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-37-207:~$ cd file
ubuntu@ip-172-31-37-207:~/file$ nano ubuntu
ubuntu@ip-172-31-37-207:~/file$ cat ubuntu
This Virtual Machine was created in aws using EC2.
ubuntu@ip-172-31-37-207:~/file$ pwd
/home/ubuntu/file
ubuntu@ip-172-31-37-207:~/file$

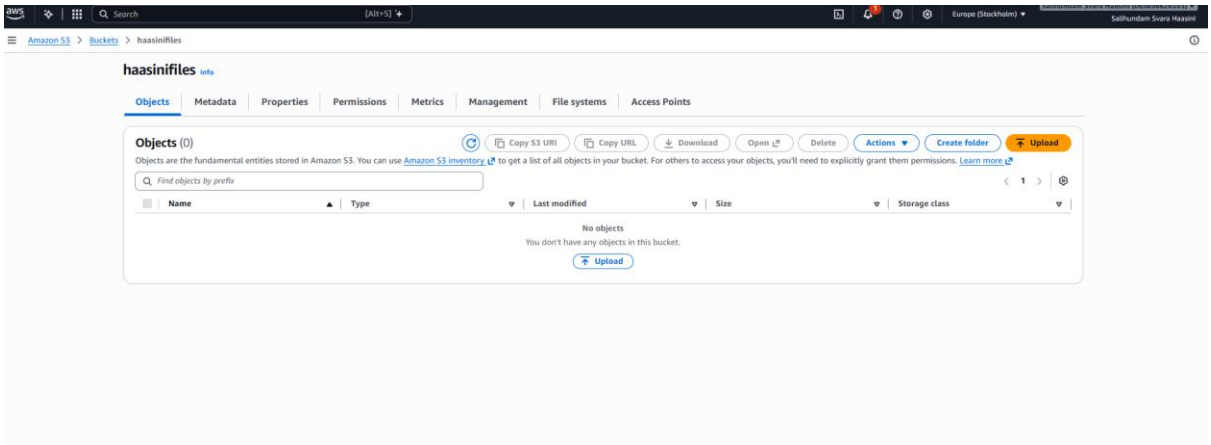


TASK : b) S3 Bucket Creation and Upload Objects

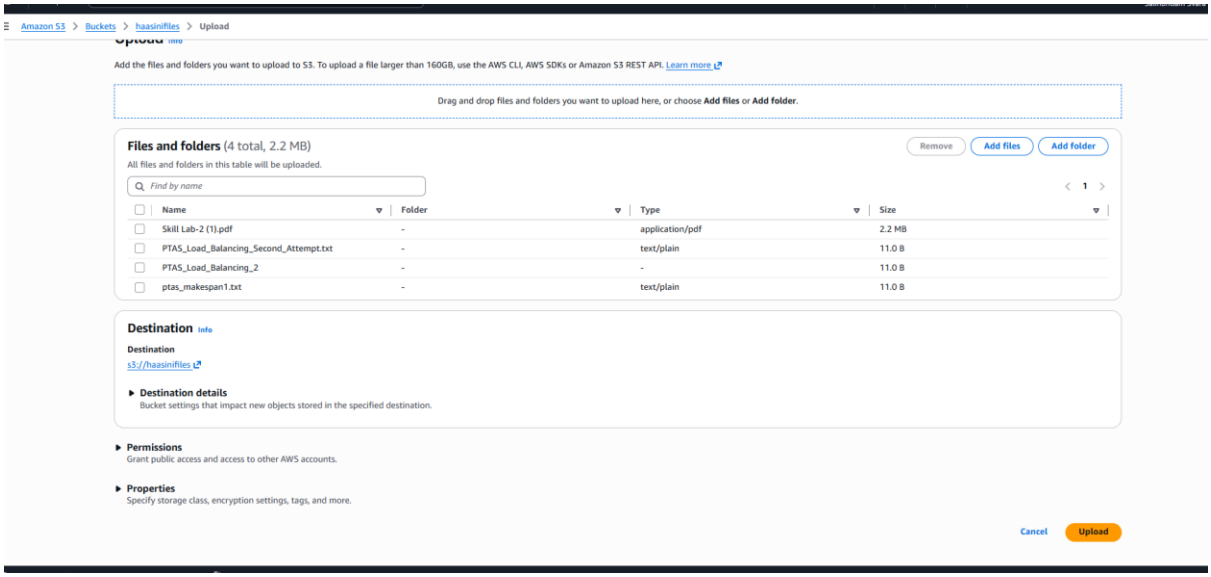
I. Create an S3 Bucket for file storage.

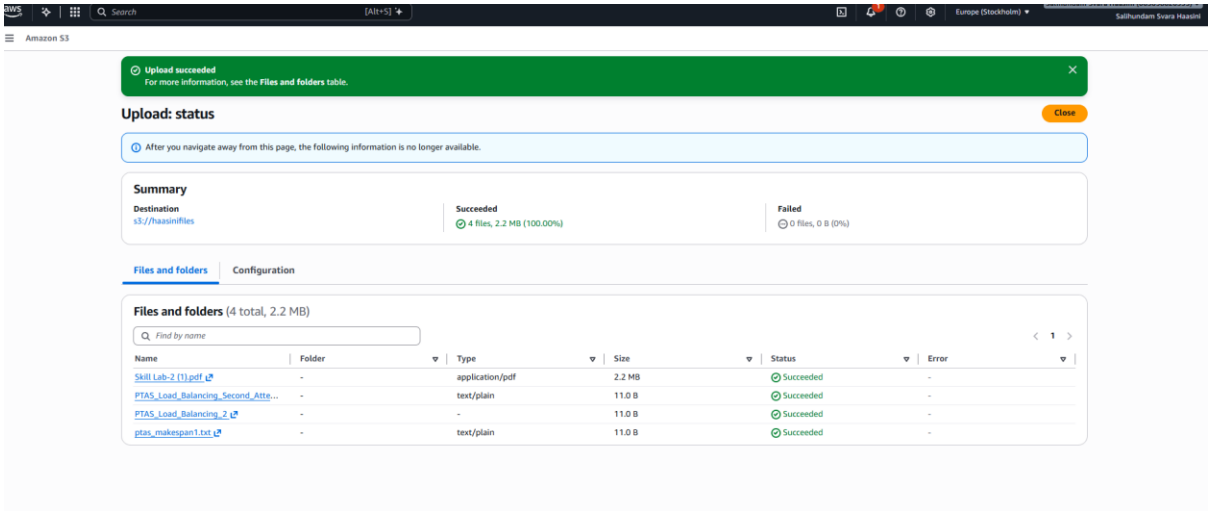






II. Upload objects with different file extensions.




Search
[Alt+S]
Europe (Stockholm)
Salihundam Svara Haasini
Amazon S3
Upload succeeded
For more information, see the Files and folders table.
Upload: status
Close
After you navigate away from this page, the following information is no longer available.
Summary
Destination
s3://haasinifiles
Succeeded
4 files, 2.2 MB (100.00%)
Failed
0 files, 0 B (0%)
Files and folders
Configuration
Files and folders (4 total, 2.2 MB)
Find by name
1
Name
Folder
Type
Size
Status
Error
Skill Lab-2 (1).pdf
-
application/pdf
2.2 MB
Succeeded
-
PTAS_Load_Balancing_Second_Atte...
-
text/plain
11.0 B
Succeeded
-
PTAS_Load_Balancing_2
-
-
11.0 B
Succeeded
-
ptas_makespan1.txt
-
text/plain
11.0 B
Succeeded
-